Collective flow of circadian clock information in honeybee colonies

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1 Abstract

Honeybee colonies exhibit a collective circadian rhythm, coordinating thousands of workers across rhythmic (e.g., nectar inflows) and non-rhythmic (e.g., brood care) tasks. However, how these rhythms integrate at the colony level remains unclear. Using an AI-driven tracking system, we continuously monitored all individuals in two colonies, revealing that circadian rhythmicity is present across all age groups but shifts by up to two hours in workers furthest from the entrance. Our data and an agent-based model suggest that mechanical interactions between bees transfer movement speed, propagating Zeitgeber information. This leads to a slow wave of activity, originating at the nest entrance and spreading inward, synchronizing colony-wide rhythms. This simple yet effective mechanism—bees bumping into each other—enables temporal coordination without direct environmental cues, ensuring sequential worker activation. Our findings highlight an elegant, self-organized strategy that governs temporal structure in a superorganism. To our knowledge, this is the first study to track the circadian rhythms of thousands of individuals simultaneously within a fully functioning colony.

2 Introduction

A molecular clock is present in nearly all living cells. For example, in the human body, all cells have a clock that is generated by a complex molecular feedback loop, involving, among others, the PER and CRY proteins [31]. Operating in rhythms allows cells to anticipate and respond to daily environmental changes, such as light and darkness. When each cell operates independently with its own unique rhythm and phase, it's likely that the complex interplay of cellular processes would become disorganized. It is no surprise, then, that circadian phases in multicellular organisms are synchronized. Within the brain, for example, the suprachiasmatic nucleus of the hypothalamus acts as a major Zeitgeber (a German term for "time giver", referring to any external or internal cue that entrains biological rhythms) for the remainder of the brain and the body [10]. Information about the daily rhythm from outside is provided by visual input from the eye, with both dedicated photoreceptors[1] and pooled information from all photoreceptors contributing information about light and dark day times. Information flow is provided by neurotransmitter release and hormonal regulations [10]. The coupling of multiple networks allows for the circadian clock in the mammalian body to remain synchronized across all tissues, with very little time lag. When this coupling is disrupted, leading to a desynchronization of peripheral and central clocks, the result leads to pathological situations and a diversity of diseases [10].

Social insects are classic examples of superorganisms: the reproductive unit is the entire colony, with hundreds to thousands of typically non-reproductive workers [14]. To operate as an integrated collective, however, the individual workers must coordinate their actions, similar to how cells unite to form a multicellular organism [30]. In colonies of the Western honey bee, *Apis mellifera*, different tasks are allocated via age polyethism and social experience: young individuals perform nursing duties, old individuals forage (to name but the two most prominent tasks) [23, 37]. Nursing duties are necessary throughout day and night, and are performed in constant darkness in the nest. In contrast, foraging duties are only possible with sufficient light during the day, and so foragers can rest at night [17, 16]. It is no surprise, then, that across the social insects, circadian rhythms have been shown in foragers, while nurses hardly show any overt behavioral circadian rhythm [7, 19, 32, 25, 26, 3, 12, 18, 9]. Indeed, nurses have no access to external light sources and thus no direct abiotic circadian Zeitgeber. However, when removed from the nest, nurses do show a synchronized circadian rhythm, showing that circadian information is present, even when not displayed strongly in their behavior [11, 25, 26, 28]. These observations raise two important questions. First, how strict is the lack of circadian rhythm across nurse bees; do nurses and foragers form two independent groups, or is there a continuum that includes other castes/tasks? Second, given that nurses remain rhythmic despite lacking direct access to the external light cycle, how does circadian information reach them: is circadian entrainment independent across individuals in the colony and regulated by an abiotic Zeitgeber, or is it coordinated using a social Zeitgeber, and if so, what is the underlying mechanism? In the absence of light within the nest, abiotic common Zeitgebers could include temperature, vibrations or odor inflow. Conversely, a social Zeitgeber would rely on collective

behaviors driving circadian synchrony. Previous data suggest that nurses are entrained by social factors such as volatile pheromones, vibrationvibrational signals from nestmates (distinct from the abiotic substrate vibrations noted above), and changes in the microenvironment such as humidity and CO2 concentration, though the details remain to be elucidated [11].

Here, we investigate the collective flow of circadian information within a honeybee colony, across all individuals. By leveraging a system that tracks thousands of bees over weeks, we provide a unique whole-colony perspective on circadian organization. Recent developments in machine learning, tracking all individuals, and data analysis allowed us to follow the individual movement patterns of thousands of bees within the colony, over long periods of time and different years, for both foragers and nurses. We found that nurses, despite their round-the-clock work and lack of light exposure, do show a prominentsignificant (but weak) circadian rhythmicity. Furthermore, we find a shift in phase along a spatial gradient within the nest, suggesting that biotic rather than abiotic factors function as dominant Zeitgeber. Finally, we show that mechanical interaction among bees is a sufficient factor to explain the observed data: bees moving and bumping into each other act as social Zeitgebers to synchronize the entire colony into a common circadian rhythm. This is, to our knowledge, the first report about a collective system that entrains a circadian rhythm by mechanical interaction generating a sizable phase shift.

3 Results

We studied the age-related and spatial organization of circadian activity within honey bee colonies using long-term automated tracking of all individuals in two observation hives. Specifically, we continuously video-recorded two queenright colonies of *Apis mellifera carnica* over a period of 57 days (colony A) and 100 days (colony B). We regularly introduced uniquely marked bees, creating colonies in which each individual was identifiable, and of known age. The bees had access to the outside world via a tube, allowing them to forage in the wild, and were otherwise left undisturbed. Videos were processed to provide continuous trajectories and IDs for all individuals in the hive [35, 4, 38, 37]. In total, we tracked 1917 bees in colony A and 3404 bees in colony B. At any given day a minimum of 716 bees were visible (max: 1320 in colony A, 1594 colony B). We calculated each bee's movement speed as the change of their position over time. Circadian rhythm was assessed for each bee by fitting a cosine with a period of 24 hours to movement speed data (Figure 1a). A direct spectral (Lomb–Scargle) estimate confirms a dominant period near 24 h (Supplementary Note 4). For each bee and each day, we used two parameters in subsequent analyses: the time of the activity peak $T(\phi)$ and the prominence of the rhythm (R^2) (Figure 1b). For bee groups we calculated synchronicity S as the inverse normalized standard deviation of the phase (see Methods).

3.1 All ages rhythmic, distance to nest entrance linked to rhythmicity, synchronicity, and phase

First, we analyzed how circadian rhythmicity changes with an individual's age. We split bees by age cohorts of 5 days each, and quantified the percentage of rhythmic bees within each group(Figure 1c). We found significant daily rhythms in all age groups, and with increasing age the proportion of rhythmic bees increased. For the youngest cohort (bees aged 1-4 days), we found 29% and 64% significantly rhythmic bees in colony A and B, respectively (for the very first day, the numbers were 18% and 58%, respectively). Conversely, more than 89% and 93% of the bees aged 35 days or more had significant circadian rhythms (Figure 1c).

Next, we investigated how age and rhythmic activity are organized spatially in the nest. To do this, we mapped the median age for each position on the comb. As expected [34, 24], we found a systematic and continuous decrease in age with increasing distance from the nest entrance (Figure 1d). This decrease in age is correlated with a delay in peak activity time, shifting by almost two hours, from 12:20 to 14:12 (right axis Figure 1e). Therefore, the further away a bee is from the nest entrance, the later her activity peaks in the day. Interestingly, rhythm prominence also decreases with increasing distance from the entrance (R^2 in Figure 1e), and bees are also less synchronized (synchronicity in Figure 1e). This spatial pattern in rhythmicity is likely due to age polyethism: young bees are nurses caring for brood in the center of the nest, middle-aged bees offload foragers at the entrance and process nectar into honey, and foragers collect resources outside and pass them off to nestmates close to the nest entrance. Therefore, the spatial organization of circadian phase and activity strength within the nest can be mapped onto bee castes: nurse bees show a weaker rhythm and a later activity peak than foragers. The continuous character of this shift (Figure 1e) reflects the continuity of cast development across several tasks within the lifespan of bees [37]. Note that the distance relationship is not strictly monotonic: in Figure 1e, the youngest workers are 300 mm from the entrance (brood area), and with even further distance age increases again. Importantly, however, peak activity time further decreases with increasing distance to the entrance even beyond the brood area.

3.2 Local interactions yield speed transfers

How do bees inside the dark nest entrain onto a circadian rhythm? If the Zeitgeber was an abiotic factor, such as light diffusing from the entrance (instantaneous), or floral odors spreading (within milliseconds), or temperature [15], we might also expect a reduced prominence in circadian rhythm with increasing distance from the entrance, but we would expect the rhythm to be synchronized. Therefore, we hypothesized that circadian entrainment could be caused by physical interaction among bees. This would create a local mechanism that spreads activity from one individual to the next [33]. To test this idea, we identified all interactions of bees by quantifying their proximity: we defined each bee as a rectangle of 14 mm by 6 mm, and extracted all pairs with overlapping

rectangles. For each interaction, we analyzed the speed change by subtracting the average speed of 30 seconds before an interaction from the speed 30 seconds after the interaction. Since this was done for each bee in a pair (every bee was a "focal bee" and an "interacting partner" once), each interaction yielded two speed-change data-points. We grouped both the focal bees and the interacting partners into six quantiles, and plotted the speed change for all pairings in a 6x6 matrix of 36 cells (Figure 2a, left panel). We found that fast bees become slower, in particular when interacting with slow bees, and slow bees increase their velocity upon interaction, in particular when interacting with fast bees (Figure 2a, left panel "Real data"). This indicates that physical interaction leads to a transfer of movement activity.

Is this movement transfer significantly different from the general speed changes in the colony that are not linked to physical interactions? To test this, we constructed a null model by sampling two bees on the comb surface randomly for every interaction time point and assessing their speed changes as before. Even in this baseline condition, fast animals tend to become slower and slow bees on average become faster (regression to the mean) (Figure 2a, right panel, "Null model"). This can be explained, in that speed values are bounded. When a bee is moving at its maximum speed, any speed change can only be a deceleration due to energy constraints and mechanical limitations. By the same logic, bees that do not move can only remain still or accelerate. However, when we statistically tested the null-model against the real data, we found a significant difference in the vast majority of cells (Welch's t-test p ¡ 0.05 in 35 out of 36 cells for colony B, 34 out of 36 cells in colony A). These proportions are robust to Benjamini–Hochberg correction and to a generalized-linear-model cosinor (Supplementary Note 3). Social interactions specifically, thus, are a major cause for movement speed change. Analogous to physical particles transferring momentum when they collide, we can think of this process as a "speed transfer" between two individuals.Unlike physical momentum which is conserved, here we observe a behavioral contagion where activity stimulates further activity. We term this process "speed transfer" between two individuals.

Grouping bees by time of peak activity $T(\phi)$, we find that bees with a late peak of activity get accelerated the most, in particular by early peaking bees which on average are older, closer to the entrance and more pronounced in their rhythm (Figure 2b, left panel). A similar observation is valid for age and R^2 (see Supplementary Information S3). Again, a comparison with shuffled data in a null model (Figure 2b, right panel) shows a high significance (36 out of 36 with p ¡ 0.01). Speed change for these groups is less pronounced as when comparing speed groups directly, suggesting that the driving force is speed itself, and not age, R^2, or $T(\phi)$. The latter show a significant effect due to the inherent correlation of speed with age and or phase ϕ.

Speed transfer across bees might be due to a non-directional arousal effect, where being touched by another bee leads to increased activity, or it could be due to a directional nudging effect. To differentiate between these alternatives, we mapped speed changes as a function of where around the focal bee's the body the interaction took place, by projecting speed change onto the ego-centric coordinate system of each interaction partner. We plotted how the focal bee changes speed depending

on where she was touched by her sister (Figure 2c). The resulting map shows that, on average, when a bee touches her sister with her head she is slowed down (blue circles in Figure 2c), while when she interacts at the sides, and more so at the back, she speeds up (yellow circles in Figure 2c). Conversely, her interaction partner becomes faster when being touched by the focal bee's head (yellow circles in Figure 2d), and slows down when being touched by the focal bee's back (blue circles in Figure 2d). Given that bees generally move forward, this result suggests a simple physical interpretation: when a bee bumps into the back of a nestmate with her head, she slows down, while her partner is accelerated. Speed transfer is a directional nudging effect. However, the situation is different for resting bees (speed ¡ 1 mm / sec): these bees accelerate with every interaction, suggesting that for resting bees an arousal effect, independent from the interaction site, is the best explanation (see Supplementary Figure S3).

3.3 Speed transfers cause a wave-like propagation of activity

The observed directional nudging effect leads to the hypothesis that young nurse bees are entrained into their circadian rhythm by the activity of older forager bees. Is this mechanism sufficient to explain the observed movement patterns in the nest? To test this hypothesis, we implemented an agent-based simulation in which two groups of agents perform random walks and transfer their movement energy when spatially interacting. Agents in the model follow a Gamma distribution for speed and Von Mises distribution for turning angles, with parameters derived from experimental data (see Methods). The two groups start off with a spatial preference for a subregion of the nest. The members of one group move at a constant speed starting at the nest center (representing the young bees deep inside the nest), the others exhibit a sinusoidal speed fluctuation and start close to the entrance of the nest (representing the old bees that are synchronized to external conditions). Social interactions transfer speed similar to the effect observed in the data, such that slow bees gain speed from interactions. Similar to atoms at higher temperatures, the distribution of the simulated agents expands further into 2-D space when increasing their speed. The virtual nest is confined: agents moving into the walls are reflected back. With increasing simulation time, the rhythmic group fills the available space and the number of interactions with the non-rhythmic group increases. Due to these interactions, the non-rhythmic agents move faster, spread out further, and, in effect, interact more with other agents. These interactions, however, take time to build up, creating a pronounced phase-shift in the activity of both groups. This phase shift depends on factors that determine how much time passes between social interactions, such as the speed of movement, how much both groups overlap initially, how far apart the two preferred locations are, and the standard deviation of the gaussian used to draw the initial locations of rhythmic and non-rhythmic agents (data not shown). The agent-based model shows a phase shift with increasing distance to the rhythmic group, replicating the situation found in the experimental bee data (for full ablation study see Supplementary Information), where the phase ϕ shifts with increasing distance to the nest entrance (Figure 3a,b). In fact, circadian activity, as a result of bumping physical interactions,

creates a wave of increased activity moving from the entrance into the inner area of the comb. The simulation shows that the speed transfer model is sufficient to explain the spreading of rhythmicity and the shift of the activity phase. It should be noted, though, that the model does not exclude the existence of additional mechanisms in a real bee colony.

If, as suggested by the simulation results, speed transfers are causing the non-rhythmic nurses to show rhythmicity, then it should be possible to track the initiating agents, when all bees are monitored over time, as is the case in our experiments. Therefore, we first identified the individuals that had an activating effect on a given young and rhythmic bee and recursively applied this procedure backwards in time. Every bee potentially had multiple activating interactions with multiple activating bees. Tracing back all activating paths, thus, yields a tree structure with the root representing a target nurse bee and a variable number of bumping bees per level that had speed-increasing interactions. The leaves of the resulting trees are the origins (driver bees) of activating chains of speed transfers ultimately entraining the focal bee at the root. We find that these driver bees exhibit a significantly earlier phase ϕ, higher rhythmicity R^2, and are older than the remaining bees across the entire nest (Brunner-Munzel test, p ¡ 0.001). This is reflected in the observation that deeper trees (with more layers) are initiated closer to the nest entrance (Figure 3c). Thus, we have shown that speed transfers travel from older bees with highly pronounced rhythms close to the nest entrance to younger, less rhythmic bees further inside the nest.

We analyzed the spatio-temporal distribution of bumping speed transfers. We plotted speed increase (color coded) against time of day (x-axis) and distance to the entrance (y-axis), and show that the bumping speed transfer is strongest around mid-day (9 am to 3 pm), and close to the entrance (distance ¡ 15 cm) (Figure 3d). Activity and speed transfers propagate like a wave, from the entrance deeper into the nest. As the wave travels inwards, the speed changes diminish gradually. This corroborates our interpretation that activity levels propagate from the entrance to the deeper areas of the nest, and that the entrance to the nest represents the origin of bumping interactions and of activating cascades eventually giving rise to the rhythmicity of the young bees in the brood nest.

4 Discussion

Prior work on circadian rhythms in foragers visiting flowers [34], or arrhythmic nurses performing round-the-clock brood care [19, 3], have examined individuals, but colonies are integrated collectives. Therefore, we examined how workers coordinate their temporal activity rhythms within the collective. Taking advantage of recently developed tracking technology [29, 38, 4, 37] we mapped the movement of all bees within the hive, without interruptions, across multiple days, and with known identity and age for each individual. We found that all bee casts display circadian activity, but nurses do so with less intensity. We also found that circadian activity is synchronized across bees through mechanical interaction, leading to a 2-hour time shift in activity peaks within the hive.

Interestingly, we did not find a clear distinction between "nurses" as a group, and "foragers" as another group. Rather, we found a continuum: the oldest bees, most of them foragers who often left the hive and were found close to the hive's entrance, had a strong circadian rhythm. The distance from the nest entrance predicts key properties of the daily rhythm: the further from the entrance, the weaker the rhythm, the lower the synchronization between bees, and the later the activity peaks (Figure 1). Thus, bees without access to external circadian cues are still synchronized to day and night cycles. This suggests that circadian behavior is not dictated by a particular task, since in that case we would have expected distinct groups of bees displaying a particular phenotype. Rather, circadian behavior is best explained by spatial location within the hive.

Our two colonies differed in size and observation length (colony A: 1,917 bees over 57 days; colony B: 3,404 bees over 100 days) but were otherwise kept under comparable conditions (queenright *Apis mellifera carnica* in observation hives with access to the outside), noting that the two were recorded in different years and seasonal windows (2016 and 2019). All central results reported here for colony B reproduce in colony A: the decline of age and rhythm strength with distance from the entrance, the interaction-driven speed transfer, and the distance-dependent phase lag (Figures S1a–S1e and S2). Quantitative differences between the colonies (e.g. absolute numbers of bees and interactions) scale with colony size and recording duration and do not affect these qualitative phenomena.

We identified cascades of mechanical speed-transferring interactions as the main mechanism driving this rhythm. Circadian timing emerges as a property coordinated via local arousals propagated throughout the nest. This behavioral mechanism exploits the individual circadian propensity of each bee: the bumping does not create a circadian rhythm, but rather only entrains the phase of the rhythm. As a result, we observed a wave of increasing activity traveling across the comb.

This wavelike phenomenon is not a transfer of mechanical energy but can be thought of as an activity wave that both reflects and affects the statistical properties of the individuals it reaches. Unlike in multicellular organisms where hormones and/or neural signals create almost instant synchrony [10], the speed transfer cascades produce activity in deeper parts of the nest with a time-lag of up to two hours. Such a delay may hold adaptive benefits for colony organization, as it aligns worker activation with resource flow: food-processing bees, located deeper within the nest, become active after foragers initiate their tasks.One consequence of such a delay is that worker activation becomes aligned with resource flow, with food-processing bees deeper in the nest becoming active after foragers initiate their tasks. We note this alignment as an observation only: our data and model do not test whether it is adaptive, and it may be a non-adaptive emergent property of colony dynamics rather than a trait selected for this function. From the perspective of coupled oscillator theory, the colony behaves as a system of weakly coupled oscillators. The observed phase lag in younger bees suggests they are more weakly coupled to the entraining signal (the foragers) than the foragers are to the external light cycle. Although increased activity is energetically costly, moving faster can enhance efficiency by increasing throughput per unit time. Additionally, the heightened activity in specific regions, such as the brood nest or honey storage areas, likely boosts task-specific

functionality. Increased movement in these zones could not only indicate elevated processing rates but also create greater permissibility for other workers, facilitating the transfer of resources and information.

Our data cannot exclude that, in addition to the behavioral 'bumping' mechanism, other Zeitgebers might contribute to synchronizing bee groups. It has been shown that social factors such as odors from the outside world brought in by foraging bees, vibrations (both from outside or caused by returning foragers dancing on the comb) [20, 27, 28]. Abiotic factors such as temperature shifts and diffusing light might also contribute, but are not necessary [28]. However, all of these factors have in common that their spatial spread is fast, if not immediate. Indeed, their effect was studied using individuals removed from the hive. Thanks to our approach to observe all bees, over long time periods, within a full hive, we were able to demonstrate a 2-hours time lag, which is a strong indication that the dominant mechanism for social entrainment is a mechanism that is slow and behavioral. Thus, we propose that mechanical bee-to-bee interaction is the main synchronizing mechanism across bees.

We note that our study uses movement speed as a proxy for circadian activity. While movement is a robust behavioral readout, it does not directly measure physiological arousal or metabolic rate. However, given the strong correlation between movement and task performance in honeybees, we consider it a reliable metric for colony-level rhythms.

It is important to distinguish between the parameters of our agent-based model and the metrics used for empirical data analysis. The model's interaction parameters were chosen to phenomenologically reproduce the wave-like spread, whereas our empirical analysis quantified physical contacts directly from tracking data. Furthermore, our model intentionally excludes other potential mechanisms (such as chemical signaling or thermal gradients) to test the sufficiency of simple mechanical interactions. We do not claim these other factors are absent, but rather that they are not strictly necessary to explain the observed phase-shift phenomenon.

The superorganism honeybee colony has often been compared to a multicellular organism. Here we show that, for the organization of circadian timing, the bee superorganism uses a slow spatial mechanism unknown in multicellular organisms. While in, say, the mammalian body circadian rhythm is synchronized using hormones and neural signals [10, 22], the behavioral bumping mechanism in bees allows for an additional level of complexity: the activity peak shifts within the hive, in accordance to the shifted need for an activity peak for each task. While foragers have to collect nectar and pollen first, receiving bees have to become active only after foragers come back from their first flight in the day. Accordingly, honey processing bees enter their chores only later. We are not aware of any cell-to-cell mechanism across neighboring cells in multicellular animals being used for synchronizing circadian activity and creating a relevant time-shift. Thus, the superorganism "honeybee colony" has evolved a new strategyexhibits a mode for circadian synchronization, unknown tothat, to our knowledge, is not known in multicellular organisms. Whether this mechanism represents an adaptation or an emergent byproduct of colony dynamics remains an open question.

Taken together, we show that a bee hive as a collective is photically entrained by external Zeitgebers sensed by the forager bees, who have access to the day/night light cycle. However, circadian rhythmicity within the hive follows a more complex pattern, whereby activity peaks are shifted thanks to the spatio-temporal organization of collective division of labor, ensuring that the different tasks in the hive are performed with an appropriate time-shift with respect to the foraging activity. A single mechanism, i.e. mechanical bumping, is sufficient to generate this remarkable organization in the beehive.

5 Methods

Beekeeping

We kept queenright colonies of *Apis mellifera carnica* in a one-frame observation hive at coordinates 52.457130, 13.296285 (Berlin, Germany). Two colonies were recorded - colony A from July 24th to September 19th in 2016 and colony B from July 4th to October 15th in 2019. Data entering our analysis was recorded between August 1st and August 25th (colony A), and between August 20th and September 14th (colony B). Colony A started with ca. 2000 bees, colony B with ca. 1500 bees. New marked bees were added throughout the experiment. The observation hive was located indoors, bees had access to the outside environment through a flexible plastic tube connecting the hive entrance to a hole in the window. A landing board was affixed at the outside to ease the bees' exit and entrance. Data in Fig. 1, 2 and 3 of the main paper relate to colony B, the corresponding data for colony A are in the Supplementary Information (section 11).

Adding marked bees

Brood from either the observation colony itself or another colony was kept in an incubator (temperature 34°C). Freshly emerged individuals were removed every day from the brood comb and individually marked at least twice a week. Bees were marked by first removing hair from the thorax using a wet toothpick, applying a thin layer of shellac, and attaching a curved, circular marker showing a binary code (for details see [35]). The number of bees marked per batch varied but never exceeded 156. A record of IDs and their respective marking dates was added to a spreadsheet. We introduced marked bees to the colony through a backdoor entrance. We prevented the emergence of unmarked bees inside the colony by exchanging the comb every 21 days. For colony A, a total of 3166 bees were marked, and throughout the data analysis period we detected 1917 unique individuals. For colony B, a total of 5099 marked bees were introduced, and we detected 3404 unique individuals during the data analysis period. Some marked bees were not accepted by the colony or might have lost their markers. Rejection and marker loss rates may vary from cohort to cohort and were not assessed quantitatively.

Bee monitoring

The two sides of the observation hive were video recorded with high-resolution cameras (2 x 12 MP per side in colony A, see [35] for details, and one 12 MP camera per side in colony B described in [37]), illuminated by synchronized infrared LED flashes (5 ms light duration) triggered by an Arduino microcontroller. Each comb side was imaged at a rate of 3 Hz for colony A and 6 Hz for colony B, alternating between both sides to avoid backlighting and ensure optimal contrast. Colony B had an additional third camera installed at the far end of the entrance tube providing a 10 Hz recording of bees as they entered or left the hive. All video data was stored to disk for analysis. We then used the BeesBook system [35, 36, 29, 38, 4, 37, 8] to automatically identify and track all marked bees over the whole duration of the recording. This yielded, for each animal and time point: unique bee ID, age, timestamp, planar coordinates, 3-D orientation. The BeesBook computer vision pipeline assigns every detection a confidence score that reflects the system's uncertainty of decoding the ID and orientation. The trajectory data was transferred to a PostgreSQL database for fast queries in our analyses. To reject false-positive detections for individuals that did not exist, we implemented a Bayesian change point model to calculate the most likely time of death for all individuals in the dataset [37]. Our marking protocol was used to filter out detections with IDs that were not yet in use.

Extracting trajectories

Sequential detections were linked to tracklets using ID information, orientation and proximity using a learned model, tracklets were linked to trajectories using a second machine learning model, as described elsewhere [4]. The trajectory-level ID and confidence were obtained by median filtering the ID and confidence values for all detections within each trajectory. Bee trajectories with confidence scores lower than 0.1 were excluded. We applied a correction procedure to correct for misalignment of certain bee tag orientation decodings relative to their body orientation as described elsewhere [8]. The resulting trajectories were used to calculate each bee's movement speed by dividing the Euclidean distance between two consecutive detections by the elapsed time (which was either $\frac{1}{3}$ s for colony A or $\frac{1}{6}$ s for colony B, or multiples thereof in case of time points with missing detections). Movement speeds greater than 15 mm/s were considered unrealistic outliers and were removed (leading to 0.1% and 0.2% removed data points in colony A and B respectively).

Circadian quantification

For each bee, we calculated circadian strength R^2 and phase ϕ using the cosinor model [13, 21, 2]. Speeds were subsampled by taking the median for each hour to reduce residual dependence. For each day the fit was calculated using a three-day window (i.e., target day including the preceding and the following day, yielding windows of 72 time points, i.e., hours). We defined the movement speed v at a time step t_i (with $i = 0 \dots 71$, and $t_i = -36 + i$) as a function of the baseline mean

speed (M), the amplitude of the oscillation (A) and the phase (ϕ) with a fixed period of $P = 24$h:

$$v(t) = M + A\cos\left(\frac{2\pi t}{P} + \phi\right) \tag{1}$$

With $x(t) = \cos\left(\frac{2\pi t}{P}\right)$ and $z(t) = \sin\left(\frac{2\pi t}{P}\right)$ as known variables, and $\beta = A\cos(\phi)$ and $\gamma = -A\sin(\phi)$ as unknown parameters, this yields:

$$v(t) = M + \beta x(t) + \gamma z(t) \tag{2}$$

We solved this linear regression problem for β and γ by using ordinary least squares. Then we deduced the parameters A and ϕ by calculating $A = \sqrt{\beta^2 + \gamma^2}$ and $\phi = \arctan\left(\frac{-\gamma}{\beta}\right) + K\pi$ where K is an integer [6]. We used an F-test [6] to test for a statistically significant amplitude greater than zero ("zero amplitude test") to identify rhythmic and non-rhythmic bees. For simplicity, in the figures we convert ϕ into the corresponding time of day $T(\phi)$ with values between 0 and 24h.

We used the prominence of the fit (R^2) to indicate the strength of the rhythm. We confirmed that R^2 serves as a good estimator for rhythm prominence by correlating it with two other measures of rhythmicity: first, we regressed it against the amplitude A ($R^2_A = 0.857$, $p < 0.001$, $r = 0.926$ for colony A and $R^2_A = 0.833$, $p < 0.001$, $r = 0.913$ for colony B), and second, against the day-night differences in bee movement speed, as difference of average movement during daytime (9:00-18:00) and night time (21:00-6:00) ($R^2_B = 0.719$, $p < 0.001$, $r = 0.848$ for colony A and $R^2_B = 0.785$, $p < 0.001$, $r = 0.886$ for colony B).

Synchronicity S, defined as the phase consistency within groups, was quantified by the standard deviation of the phase (ϕ) values across the respective individuals, normalized to [0,1] with $S = 1$ corresponding to the 5%-quantile and $S = 0$ corresponding to the 95%-quantile.

Physical interactions

We defined two bees to interact when (1) they were detected simultaneously in the hive with a confidence greater than 0.25, (2) the distance between their thorax markings was less than or equal to 14 mm, and (3) their body poses were classified as "touching". To define touching, we modeled both interaction partners as rectangles of 14×6 mm (approximately the size of a bee), centered on the bee and aligned with her body axis. The masks were then checked for overlap using a logical AND operation. Each interaction was counted as a single event. Since interactions can span multiple frames, subsequent interaction detections between a given pair were considered to belong to the same interaction event if the gaps between them were shorter than one second. For each detected interaction, we recorded the position within the nest, the timestamp, relative angle of the two bees, and the overlap area (at the start of the interaction) as well as the duration of the interaction. We treated each individual as a focal bee, yielding two data points for each interaction. For each focal bee, we recorded: relative touching positions in the focal bee's frame of reference, age of the bee, rhythmicity information (ϕ and R^2), speed, and change in speed after the interaction. The speed

change was calculated as the average speed in a 30-second window after the interaction minus the average speed during the 30-second period before the interaction started.

Statistical analysis of physical interactions

To assess whether interactions produce speed changes that are significantly different from non-interacting bees, we defined a null model. For each real interaction, we selected two random bees present in the nest at the same time points and calculated the speed changes resulting from these (fictive) interactions. We analyzed median speed change, the movement speed at the start of the interaction, the phase, and prominence of rhythm. For each of these, we divided the focal and non-focal bees into six quantiles resulting in 36 combinations. Measured interactions and the null model were compared using Welch's t-test.

Body-centered physical interactions

To quantify speed change as a function of the relative touching positions we defined an area of 16×8 mm representing the focal bee (plus 1 mm padding) and discretized this area into 16×8 bins (yielding a spatial resolution of 1 mm / bin). For every given interaction, the speed change values of both the focal bee and her interaction partner were accumulated in the focal bee's bins corresponding to its intersection with the non-focal bee's rectangle. This yields two maps representing the average speed change of focal and non-focal bee, respectively, in the coordinate frame of the focal bee. Fine-grained analysis included segmenting the data for different combinations of criteria, such as duration of the interaction or characteristics of the participating bees, e.g. age.

Agent-based modeling

We created an agent-based simulation to study the effects of speed transfers from circadian agents to non-circadian ones. The spatial arrangement mirrored the observation hive: a rectangular, two-dimensional plane bounded by walls on each side. We defined two agent populations: one representing older foraging bees, clustered at one side of the nest, and with a prominent circadian rhythm, the second representing younger nurse bees, concentrated in the middle of the nest, and without a circadian rhythm. Each agent was modeled as a point with a given 2-D position and orientation. For every simulation step, agents moved into the direction they were facing. The amount of movement (i.e., the speed) was determined by three components. Every bee received a baseline speed drawn from a Gaussian random distribution (clipped to 0 to avoid negative speeds). The forager group received an additional sinusoidal driver representing periodic speed changes. For interacting bees the slower bee received the speed difference with the faster bee. Interactions were determined by thresholding inter-individual distance. After taking a step, the orientation of each agent was changed by adding a Gaussian random sample and additionally both groups were pulled back towards the center of their respective initial distributions. Agents that reached the boundary of the arena were

reflected back. The resulting motion data was analyzed with the same procedure as the biological observations.

We created an agent-based simulation to study the effects of speed transfers from circadian agents to non-circadian ones. The spatial arrangement mirrored the observation hive: a rectangular, two-dimensional plane bounded by walls on each side. We defined two agent populations: one representing older foraging bees, clustered at one side of the nest, and with a prominent circadian rhythm, the second representing younger nurse bees, concentrated in the middle of the nest, and without a circadian rhythm.

Each agent i at time step t is described by position $\mathbf{x}_i^{(t)}$, heading $\theta_i^{(t)}$, and speed $s_i^{(t)}$. Forward speeds are drawn from a Gamma distribution:

$$v_i \sim \text{Gamma}(k, \theta) \tag{3}$$

where parameters cover the group-specific mean and variance (μ_g, σ_g) derived from data. Heading changes are drawn from a Von Mises distribution:

$$\delta\theta_i \sim \text{VonMises}(\mu = 0, \kappa = 1/\sigma^2) \tag{4}$$

For interacting pairs (i, j) within distance $d_{\text{int}} \approx 1.4$ mm, where $s_j > s_i$, the slower bee receives a speed boost:

$$\Delta s_i^{(\tau+1)} = \gamma\, \Delta s_i^{(\tau)} + \max(s_j^{(\tau)} - s_i^{(\tau)}, 0) \tag{5}$$

where γ is the decay rate. Agents are also attracted toward their home location μ_g (homing pull).

Parameters used in the simulation are summarized in Table 1.

Table 1: Model Parameters

Parameter	Value	Unit
Comb size (L_x, L_y)	354×205	mm
Speed scale σ_s	50	-
Interaction threshold d_{int}	1.41	mm
Decay rate γ	0.8	-
Homing pull h	0.1	-

The resulting motion data was analyzed with the same procedure as the biological observations.

Speed transfer cascade trees

Binary interactions create a tree-like interaction history that can be traced, such as bee a being activated by bee b1 and bee b2, which in turn are activated by bees c1..cn, and so on. Interactions were only considered between 10:00 and 15:00. We selected a random subset of $n = 1000$ bees from those that were significantly rhythmic, younger than 5 days old, and reached their peak activity

after 12:00. These bees were taken as our target group for the ends of interaction chains. We then identified, for each bee, all interactions that caused a positive change in speed, and traced back interactions in time. This yielded a tree structure, where the root vertex was fixed to be a young rhythmic bee and all bees that had activated her were added in the previous timestep. The maximum time window between two interactions was set to 30 minutes, and the maximum duration of a cascade was set to be 2 hours. After creating such trees, we extracted all paths from the root back to the leaf nodes. To test if phase ϕ, R^2, or age differed between the leaf nodes and the general hive we used a Brunner-Munzel test [5].

6 Data Availability

The raw data (video files, trajectories) are approximately 400 TB in size and available from the corresponding author on reasonable request. All derived datasets (velocities, cosinor fits, interactions and interaction trees) analysed during the current study are available online: https://zenodo.org/records/10869728.

7 Code Availability

All code is available online: https://github.com/BioroboticsLab/speedtransfer.

8 Author Contributions

Conceptualization: J.M., W.K., B.W., D.M.D., A.Z., M.L.S., C.G.G. and T.L.; Methodology: J.M., W.K., B.W., D.M.D., A.Z. and T.L.; Software: J.M., W.K., B.W., D.M.D., A.Z. and T.L.; Resources, supervision: T.L.; Project administration: T.L.; Data curation: J.M., W.K., B.W. and D.M.D.; Writing: J.M., W.K., B.W., D.M.D., M.L.S., C.G.G and T.L; Visualization: J.M., W.K., B.W. and D.M.D.

9 Figures

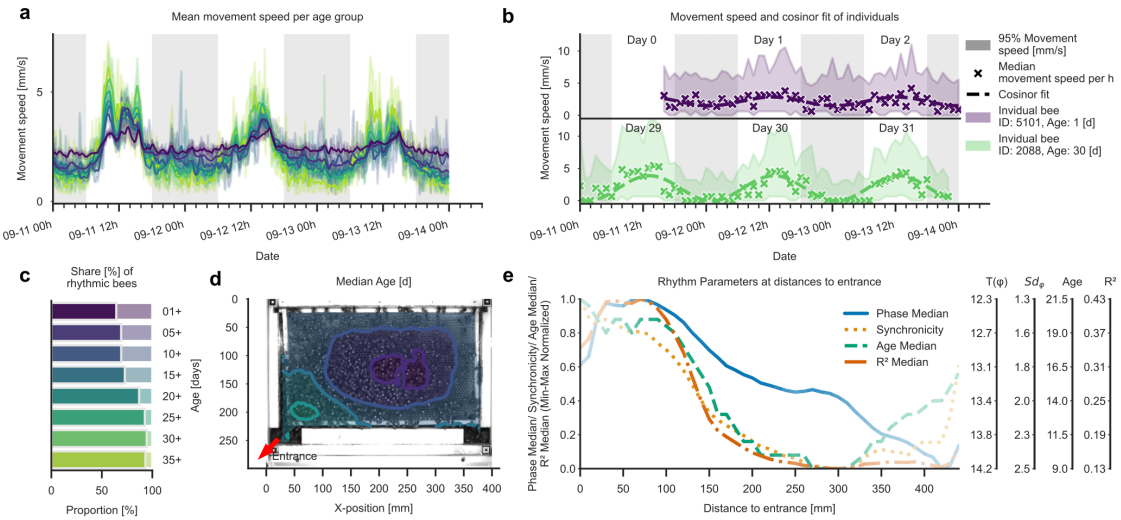


Figure 1: **Circadian rhythms across bee age and nest space (a)** All age groups show higher speeds during the day than at night, but the difference increases with bee age. Example movement speed [mm/s] of bees for three days, split by age from young (purple) to old (yellow). Data was binned to hours, and smoothed with a Gaussian ($s_{\text{smoothed}}(t) = \sum_{u=-z}^{z} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{u^2}{2\sigma^2}} s(t-u), \sigma = 4, z = 33$). Lighter bands show 95% confidence intervals of the age group unsmoothed velocities. **(b)** Fitting a cosine function allows to extract circadian activity parameters. Two individuals are shown as example: a young nurse (top, purple) and an older forager (bottom, green), each for a three-day window. The cosinor fit is significant for both animals, but amplitude and variability are higher for the nurse bee. Note that some time points are missing for the young bee on day 0, since it was freshly introduced to the observation hive. **(c)** With increasing age, the share of significantly circadian bees increases. Bees are split in age groups as in (a), and the proportion of significantly rhythmic bees is shown for each group. **(d)** Spatial distribution of median age over the comb. Older bees (foragers) tend to be closer to the entrance, while younger bees (nurses) are more concentrated in the center of the comb. **(e)** Circadian parameters decrease with increasing distance from the entrance. Curves indicate peak phase P (daytime of maximum), phase standard deviation (hours), and circadian prominence R^2 as calculated with the cosinor fit, relative to distance to entrance. Bee age (days) also decreases with increasing distance from the entrance. Curve color saturation indicates data density (i.e., how many bees were in that area). Synchronicity is defined as the standard deviation of the phase. The four scales to the right correspond to the original values for the four normalized traces. Data shown for colony B. For colony A refer to SI Fig. S1.

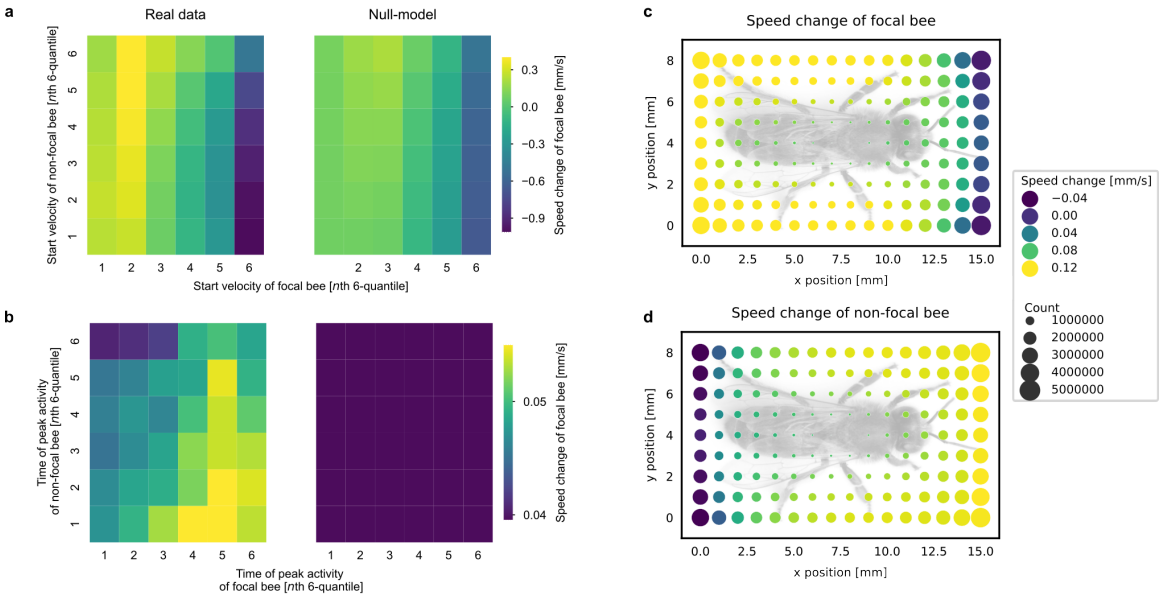


Figure 2: **Dynamic physical interaction leads to speed transfer across bees (a)** Fast bees are slowed down when bumping into slow bees. Focal bees (x-axis) and non-focal bees (y-axis) were sorted into 6 speed quantiles. For each combination, the speed change of the focal bee was color coded: speed decreases (blue hues) were particularly prominent for previously fast bees (6th quantile). The right panel shows data from a randomly shuffled null-model (see methods), experimental data is significantly different from the shuffled distribution (see Supplementary Section 11.2 for p-values). **(b)** Circadian phase ϕ of the interaction partner influences speed transfer. Focal and non-focal bees were binned in 6 quantiles depending on the phase of their cosinor fitted circadian activities. Focal bees with late phase led to higher increases in non-focal bees with early phase, and focal bees with early phase led to small increases in non-focal bees with late phase. The randomly shuffled null-model (right panel) showed no phase effect. **(c)** The plot shows speed changes of the focal bee relative to the interaction's location along her body. Speed changes resulting from interacting at the front are negative (blue hue), interactions at the back are positive (yellow hue). Since bees move forward, on average, this result indicates that bees are slowed down when bumping into another bee, and get accelerated when being bumped from the back. **(d)** This plot shows speed changes of the non-focal bee in the coordinate frame of thefocal bee. Speed changes are negative along the back of the focal bee, and positive along her front. This indicates that bees are slowed down when bumping into the back of another bee, and accelerated when being run into. In (c) and (d), circle sizes indicate the number of observations in each point.

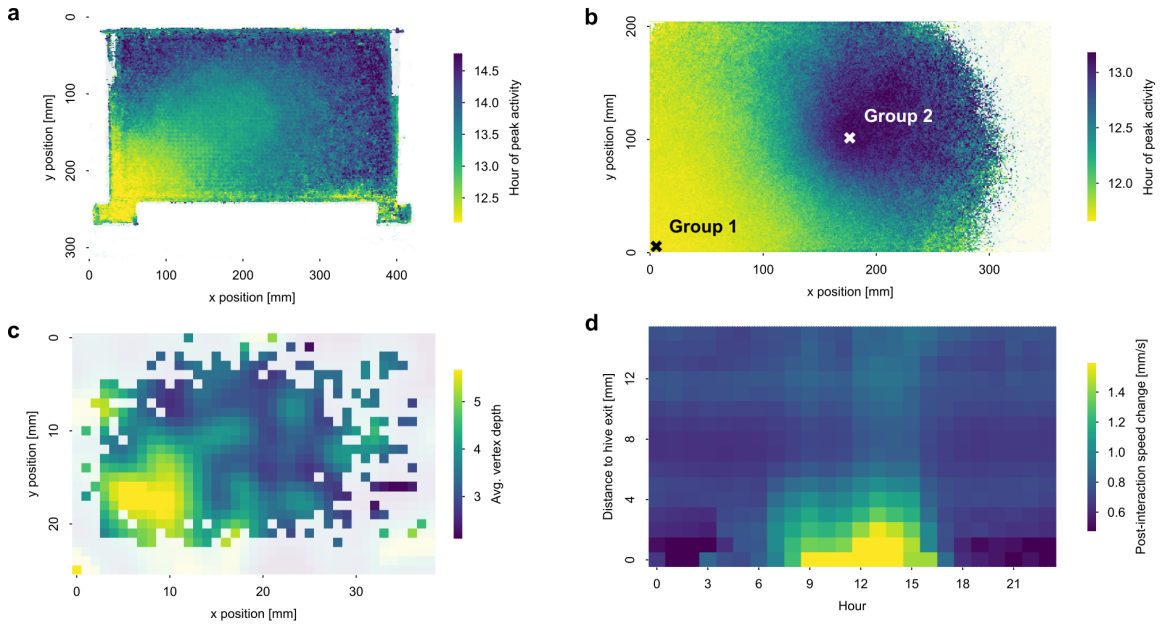


Figure 3: **Activity propagates in a wave-like manner through the nest (a)** Peak activity time (phase ϕ) is early at the nest's entrance, and increasingly late with distance to the entrance. Median phase values were color-coded onto a view on the comb. Entrance is at the lower left. **(b)** The agent-based model replicates the same spatial pattern of spatial phase distribution. Peak activity time (phase ϕ) was color-coded with a view on the simulated comb for the agent-based model. Crosses show centers of mass at the beginning of the simulation run for group 1 (initially rhythmic, lower left) and group 2 (initially non-rhythmic). **(c)** Bees initiating interaction trees are mostly localized at the nest entrance. The plot shows the average depth of interaction tree vertices at each nest location. The larger the depth, the more interactions will be downstream of that bee. **(d)** Largest interaction effects are localized at the entrance, and occur around noon. This two-dimensional false-color coded plot shows speed increase resulting from an interaction as a function of time-of-day (x-axis) and distance to the entrance (y-axis). The deeper into the nest, the weaker and later we found speed transfers.

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11 Supplementary Information

Figure S1: Relates to Fig. 1. Circadian rhythms across bee age and nest space for colony A. While Fig. 1 in the main manuscript relates to colony B, here we show the same analysis for colony A.

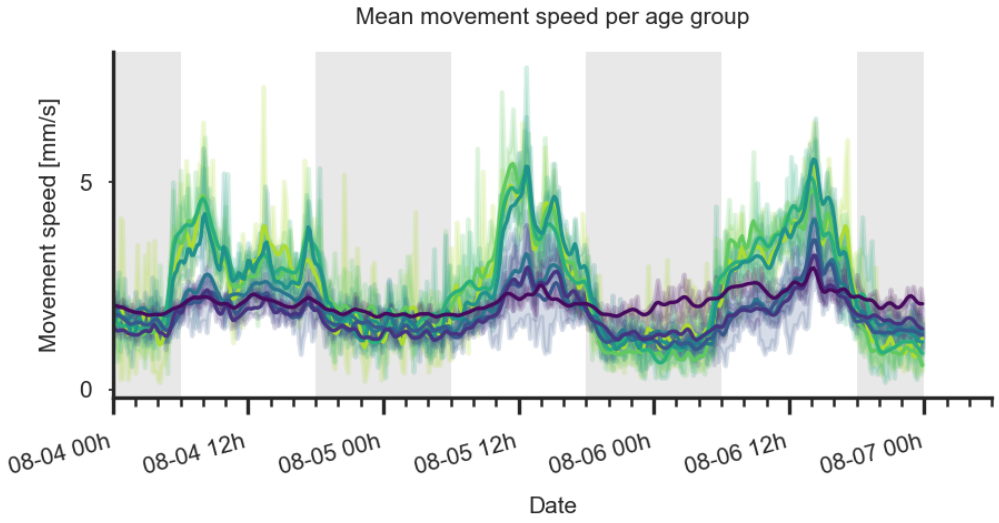


Figure S1a: All age groups show higher speeds during the day than at night, but the difference increases with bee age. Example movement speed [mm/s] of bees for three days, split by age from young (purple) to old (yellow). Data was binned to hours, and smoothed with a Gaussian ($s_{\text{smoothed}}(t) = \sum_{u=-z}^{z} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{u^2}{2\sigma^2}} s(t-u), \sigma = 4, z = 33$). Lighter bands show 95% confidence intervals of the age group unsmoothed velocities.

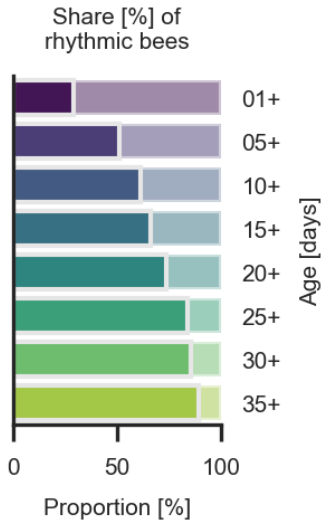


Figure S1c: With increasing age, the share of significantly circadian bees increases. Bees are split in age groups as in (a), and the proportion of significantly rhythmic bees is shown for each group.

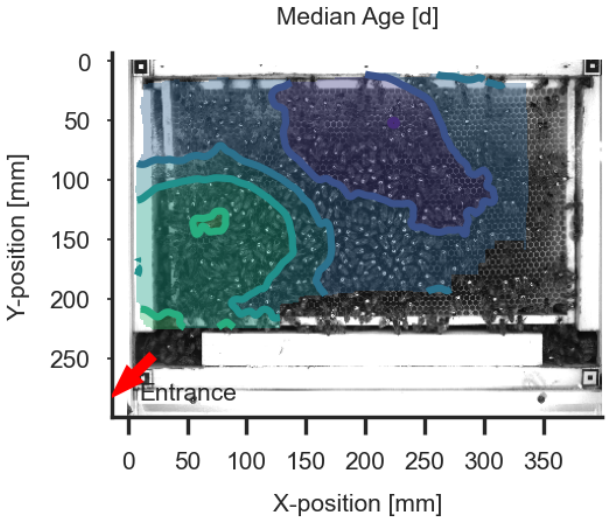


Figure S1d: Spatial distribution of median age over the comb. Older bees (foragers) tend to be closer to the entrance, while younger bees (nurses) are more concentrated in the center of the comb.

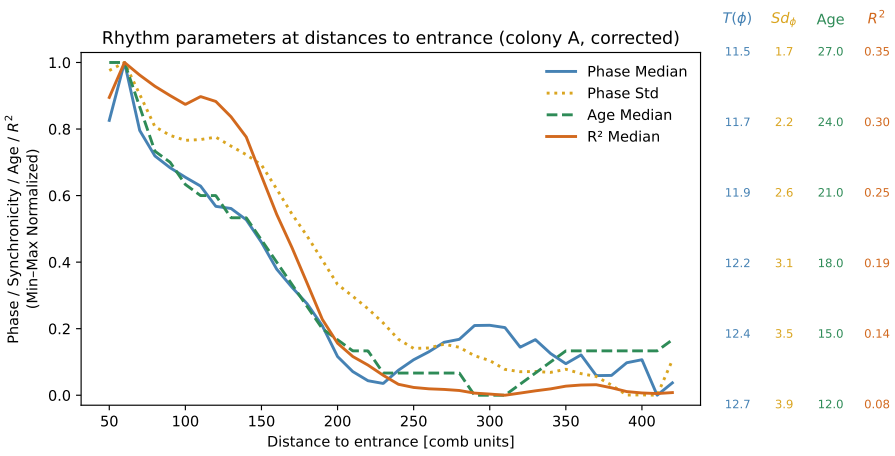


Figure S1e: Circadian parameters decrease with increasing distance from the entrance. Curves indicate peak phase P (daytime of maximum), phase standard deviation (hours), and circadian prominence R^2 as calculated with the cosinor fit, relative to distance to entrance. Bee age (days) also decreases with increasing distance from the entrance. Synchronicity is defined as the standard deviation of the phase. The four scales to the right correspond to the original values for the four normalized traces.

11.1 Figure S2. Relates to Fig. 2. Dynamic physical interaction leads to speed transfer across bees (colony A). While Fig. 2 in the main manuscript relates to colony B, here we show the same analysis for colony A.

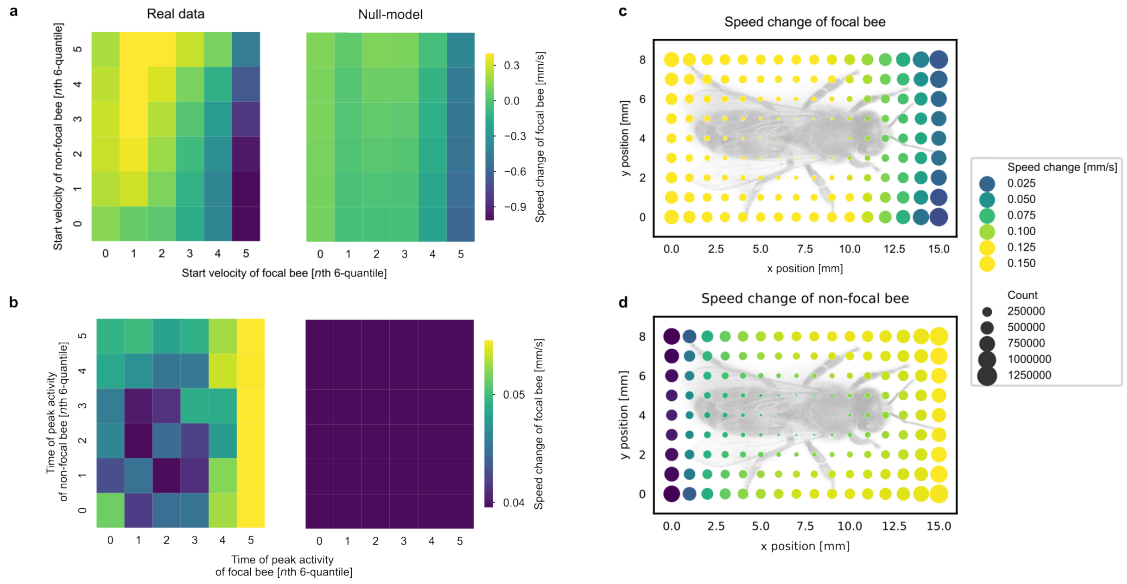


Figure S2: **(a)** Fast bees are slowed down when bumping into slow bees. Focal bees (x-axis) and non-focal bees (y-axis) were sorted into 6 speed quantiles. For each combination, the speed change of the focal bee was color coded: speed decreases (blue hues) were particularly prominent for previously fast bees (6th quantile). The right panel shows data from a randomly shuffled null-model (see methods), experimental data is significantly different from the shuffled distribution (see Supplementary Note 2 for statistical values). **(b)** Circadian phase ϕ of the interaction partner influences speed transfer. Focal and non-focal bees were binned in 6 quantiles depending on the phase of their cosinor fitted circadian activities. Focal bees with late phase led to higher increases in non-focal bees. The randomly shuffled null-model (right panel) showed no phase effect. **(c)** Bees are slowed down when bumping into another bee from the front, and accelerated when bumped from the back. Image shows speed change of the focal bee (shown schematically in grey) relative to the interaction location along her body. Speed changes resulting from interacting at the front are negative (blue hue), interactions at the back are positive (yellow hue). **(d)** Bees are slowed down when bumping into the back of another bee, and accelerated when hit by a head. Image shows speed change of the non-focal bee relative to the interaction location along the body of the focal bee (shown schematically in grey). Speed changes are negative along the back of the focal bee, and positive along her front. In (c) and (d), circle sizes indicate the number of observations in each point.

11.2 Local interactions yield speed transfer: P-values for testing interaction metrics against, non-interaction null-model

For any of the two colonies, focal bees (columns) and non-focal bees (rows) of all interactions were grouped into six quantiles of four features (speed at interaction start, phase, R^2, age). For each combination, the change in velocity of the focal bee was compared to a null model (see Methods) using a Welch test. The tables show the corresponding p-values. Tests were performed on independent subsets of data. Tables 1-4 relate to colony B, tables 5-8 relate to colony A.

Supplementary Table 2: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective interaction start speed quantiles (colony B). Relates to Fig. 2a.

	1	2	3	4	5	6
1	0.0	0.0	0.0	0.0	0.0	8.52e-06
2	4.33e-22	1.11e-176	0.0	3.39e-244	0.02	0.0
3	6.09e-17	3.57e-74	1.90e-181	1.56e-44	7.29e-38	0.0
4	0.81	1.03e-16	2.18e-08	1.11e-08	3.16e-154	0.0
5	0.0	0.0	0.0	0.0	0.0	0.0
6	0.0	0.0	0.0	0.0	0.0	0.0

Supplementary Table 3: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective phase quantiles (colony B). Relates to Fig. 2b.

	1	2	3	4	5	6
1	4.42e-91	3.94e-64	3.84e-71	5.82e-76	9.68e-52	9.39e-168
2	2.35e-125	2.53e-158	9.99e-180	4.27e-248	1.77e-298	3.44e-134
3	2.51e-135	5.50e-149	3.05e-180	1.86e-290	7.84e-214	7.56e-136
4	1.39e-122	7.62e-118	7.88e-151	3.44e-147	7.15e-126	1.23e-137
5	2.53e-122	8.93e-139	1.26e-81	3.61e-103	2.47e-71	2.38e-134
6	3.48e-150	2.66e-107	2.41e-92	1.28e-73	8.88e-49	3.82e-120

Supplementary Table 4: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective R^2 quantiles (colony B).

	1	2	3	4	5	6
1	6.23e-06	0.00	9.11e-16	4.81e-23	9.06e-18	5.44e-96
2	1.48e-70	8.26e-102	5.68e-138	9.40e-133	1.96e-152	4.33e-147
3	4.32e-153	2.48e-190	2.53e-241	5.20e-243	2.25e-206	6.54e-135
4	3.24e-187	1.20e-196	5.28e-247	2.56e-253	2.06e-183	1.52e-150
5	2.69e-169	1.71e-209	2.24e-205	4.03e-191	3.69e-130	7.03e-133
6	9.45e-173	1.08e-185	1.14e-168	5.11e-145	4.35e-135	1.46e-98

Supplementary Table 5: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective age quantiles (colony B).

	1	2	3	4	5	6
1	4.11e-67	7.07e-47	4.92e-25	1.89e-13	1.37e-18	5.69e-47
2	8.55e-152	8.63e-121	1.57e-84	1.06e-69	2.62e-77	3.85e-58
3	2.70e-180	6.01e-184	4.82e-184	6.26e-131	6.95e-65	4.58e-56
4	2.36e-258	1.06e-271	2.19e-255	7.13e-121	1.82e-81	3.86e-75
5	0.0	0.0	4.09e-151	7.35e-84	6.79e-61	2.97e-58
6	0.0	0.0	1.86e-80	5.60e-39	4.08e-35	1.76e-17

Supplementary Table 6: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective interaction start speed quantiles (colony A).

	1	2	3	4	5	6
1	9.61e-192	0.0	1.12e-300	3.85e-309	5.21e-284	0.0
2	0.0	5.07e-137	1.82e-18	0.10	0.15	0.05
3	2.64e-157	1.15e-134	1.44e-79	1.62e-124	5.86e-122	1.40e-70
4	3.78e-299	0.0	0.0	0.0	0.0	0.0
5	0.0	0.0	0.0	0.0	0.0	0.0
6	0.0	0.0	0.0	0.0	0.0	0.0

Supplementary Table 7: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective phase quantiles (colony A).

	1	2	3	4	5	6
1	1.66e-86	3.68e-74	7.92e-76	3.21e-67	1.91e-105	4.88e-181
2	1.52e-94	8.81e-64	6.92e-82	1.18e-53	2.36e-160	3.81e-75
3	3.21e-89	6.05e-82	4.20e-121	4.62e-149	1.86e-95	2.22e-112
4	3.99e-117	6.72e-121	7.86e-222	1.27e-117	7.19e-118	2.46e-97
5	1.21e-117	4.55e-161	2.20e-116	5.84e-82	8.32e-104	4.91e-74
6	6.94e-209	6.72e-95	3.72e-110	1.90e-66	1.18e-135	5.92e-76

Supplementary Table 8: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective R^2 quantiles (colony A).

	1	2	3	4	5	6
1	1.32e-23	6.48e-34	1.58e-37	3.70e-54	2.58e-123	5.85e-282
2	3.00e-81	1.19e-90	1.55e-94	4.80e-109	2.56e-205	2.87e-148
3	2.23e-102	1.17e-101	6.44e-116	3.43e-188	1.77e-167	2.05e-92
4	3.96e-93	5.44e-100	2.36e-171	1.51e-130	5.61e-128	1.08e-69
5	5.14e-102	2.68e-157	8.98e-99	1.52e-114	6.64e-95	1.56e-77
6	3.05e-152	1.70e-94	4.49e-91	1.93e-85	1.15e-73	1.82e-59

Supplementary Table 9: P-values of Welch tests comparing the speed changes of bees grouped into 6×6 pairings reflecting their respective age quantiles (colony A).

	1	2	3	4	5	6
1	3.69e-37	3.43e-70	1.42e-85	6.38e-78	4.35e-120	3.12e-266
2	1.91e-76	1.16e-69	5.82e-72	1.15e-55	1.21e-129	6.208e-127
3	2.19e-102	9.80e-87	1.41e-51	1.78e-147	1.38e-64	6.25e-112
4	3.41e-104	1.94e-73	1.64e-175	3.70e-41	3.38e-70	1.30e-73
5	6.30e-206	0.0	3.47e-56	2.58e-66	1.33e-52	4.00e-58
6	0.0	5.10e-152	1.46e-45	1.29e-35	3.57e-19	5.09e-46

11.3 Speed transferred depends on attributes of interacting bees

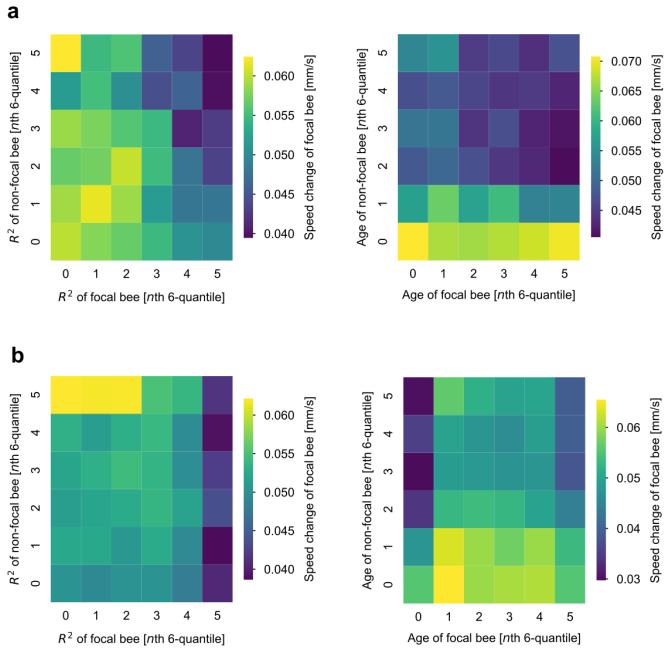


Figure S3: Average speed change of focal bee for respective combinations of 6×6 quantiles of R^2 and age for colony A (row a) and colony B (row b), respectively.

11.4 Resting bees: speed transferred irrespective of where focal bee was touched

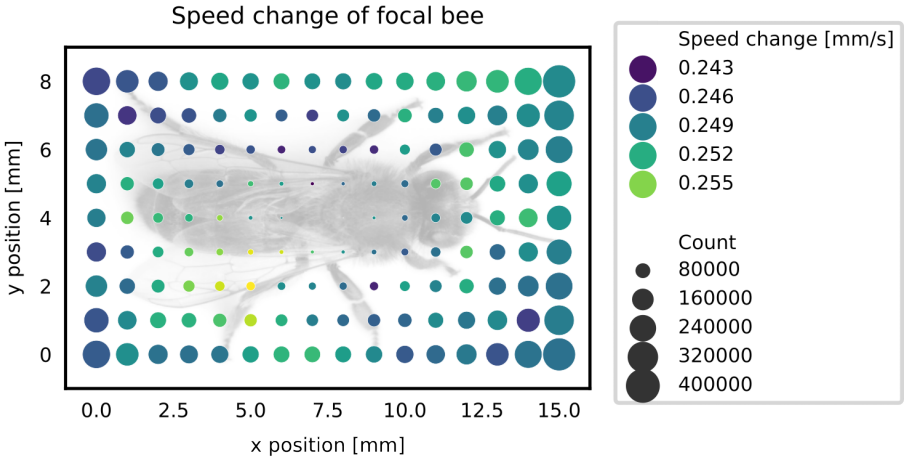


Figure S3: Resting bees are activated irrespective of where they touch their interaction partner. The average speed changes after an interaction for bins of 1×1 mm show positive values consistently around the focal bee's body.

11.5 Supplementary Note 1: Agent-Based Model Equations

Overview

The simulation models $N_1 + N_2$ agents on a two-dimensional rectangular arena representing the comb surface. Group 1 ("rhythmic" / foragers) receives an external sinusoidal speed driver; Group 2 ("non-rhythmic" / hive bees) does not. Speed transfer occurs when agents interact within a spatial proximity threshold, allowing the rhythmic pattern to propagate from Group 1 to Group 2.

State Variables

Each agent i at time step t is described by:

- Position $\mathbf{x}_i^{(t)} = (x_i, y_i)^{(t)} \in \mathbb{R}^2$
- Heading $\theta_i^{(t)} \in [0, 2\pi)$
- Accumulated speed transfer $\Delta s_i^{(t)} \geq 0$

Initialization ($t = 0$)

Positions are drawn from group-specific bivariate Gaussians centred at μ_g with covariance $\Sigma_g = \sigma_s \, \mathbf{I}_2$:

$$\mathbf{x}_i^{(0)} \sim \mathcal{N}\big(\mu_g,\ \Sigma_g\big), \qquad \theta_i^{(0)} \sim \text{Uniform}(0,\ 2\pi), \qquad \Delta s_i^{(0)} = 0. \tag{6}$$

Per-Step Update ($t \to t + 1$)

At every time step t (with integer time label $\tau = t + 1$), agents are updated in the following order.

External Driver (Group 1 only) Group 1 receives a sinusoidal speed boost that peaks once per simulated day of duration T_{day}:

$$e^{(\tau)} = \frac{\sigma_s \, \mu_1^{\text{fwd}}}{2} \left(\sin\left(\frac{2\pi \, \tau}{T_{\text{day}}} \right) + 1 \right), \tag{7}$$

where σ_s is the speed scale and μ_1^{fwd} is Group 1's mean forward speed. Group 2 receives $e^{(\tau)} = 0$.

Forward Speed Sampling Forward speeds are drawn from a Gamma distribution parameterised by the group-specific mean μ_g^{fwd} and standard deviation σ_g^{fwd}, scaled by σ_s:

$$v_i^{\text{raw}} \sim \text{Gamma}\left(k = \frac{(\sigma_s \, \mu_g^{\text{fwd}})^2}{(\sigma_s \, \sigma_g^{\text{fwd}})^2},\ \theta = \frac{(\sigma_s \, \sigma_g^{\text{fwd}})^2}{\sigma_s \, \mu_g^{\text{fwd}}} \right), \qquad v_i^{\text{fwd}} = \max\big(v_{\min},\ v_i^{\text{raw}}\big), \tag{8}$$

where $v_{\min}$ is a per-group lower-bound on forward speed and (k, θ) are the shape and scale of the Gamma distribution. The total instantaneous speed is:

$$s_i^{(\tau)} = v_i^{\text{fwd}} + e_i^{(\tau)} + \Delta s_i^{(\tau)}. \tag{9}$$

Turning Noise Heading changes are drawn from a Von Mises distribution:

$$\delta\theta_i \sim \text{VonMises}\left(\mu = \mu_g^{\text{turn}},\ \kappa = \frac{1}{(\sigma_g^{\text{turn}})^2}\right). \tag{10}$$

Velocity and Position Update The velocity vector combines the speed with the current heading:

$$\mathbf{v}_i = s_i^{(\tau)} \begin{pmatrix} \cos\theta_i^{(t)} \\ \sin\theta_i^{(t)} \end{pmatrix}. \tag{11}$$

Homing pull. Each group is attracted toward its home location μ_g:

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \sigma_s\, h\, \mu_g^{\text{fwd}}\, \frac{\mu_g - \mathbf{x}_i^{(t)}}{\left\|\mu_g - \mathbf{x}_i^{(t)}\right\|}, \tag{12}$$

where h is the homing pull coefficient.

The state is then updated:

$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} + \mathbf{v}_i, \tag{13}$$

$$\theta_i^{(t+1)} = \theta_i^{(t)} + \delta\theta_i. \tag{14}$$

Wall Reflection The arena is bounded by a rectangle $[-L_x/2,\, L_x/2] \times [-L_y/2,\, L_y/2]$. When an agent crosses a boundary, its position is clamped to the boundary and its heading is redirected toward the arena centre:

$$\text{if } x_i^{(t+1)} < -\tfrac{L_x}{2} \text{ or } x_i^{(t+1)} > \tfrac{L_x}{2}: \qquad \theta_i^{(t+1)} \leftarrow \arctan 2\big(-y_i^{(t+1)},\, -x_i^{(t+1)}\big). \tag{15}$$

An analogous rule applies along the y-axis.

Spatial Interactions and Speed Transfer At each time step, agent pairs within Euclidean distance $d_{\text{int}} = \sqrt{\sigma_s}\ \rho$ are identified using spatial hashing. The interaction threshold factor ρ sets the interaction range as a fraction of the characteristic spatial scale $\sqrt{\sigma_s}$. With the default value $\rho = 0.2$, agents interact within $d_{\text{int}} \approx 1.4$ mm, corresponding to physical body contacts between bees on the comb surface.

For each interacting pair (i, j) where $s_j > s_i$, the speed difference is computed and the slower

agent receives a boost:

$$\Delta s_i^{(\tau+1)} = \gamma\, \Delta s_i^{(\tau)} + \max_{j \in \mathcal{N}_i} \left(s_j^{(\tau)} - s_i^{(\tau)} \right)^+, \tag{16}$$

where $\gamma \in [0, 1)$ is the exponential decay rate, $\mathcal{N}_i = \{j : \|\mathbf{x}_j - \mathbf{x}_i\| < d_{\text{int}}\}$ is the set of neighbours, and $(\cdot)^+ = \max(\cdot, 0)$.

Rhythmicity Analysis

After the simulation, the mean speed across agents within each group is computed for every time step. A sinusoidal model is fitted to these time series by ordinary least squares:

$$\bar{s}_g(\tau) \approx \beta_0 + \beta_1 \sin\left(\frac{2\pi\,\tau}{T_{\text{day}}}\right) + \beta_2 \cos\left(\frac{2\pi\,\tau}{T_{\text{day}}}\right), \tag{17}$$

yielding amplitude $A = \sqrt{\beta_1^2 + \beta_2^2}$ and phase $\phi = \arctan 2(-\beta_2, \beta_1)$. The phase shift between groups is $\Delta\phi = \phi_2 - \phi_1$.

Statistical significance is assessed via a permutation test: the time labels are shuffled n_{perm} times, sine fits are recomputed, and the p-value is the fraction of permuted amplitudes exceeding the observed amplitude.

Parameter Table

11.6 Ablation Study Results

We systematically ablate individual model components to identify which mechanisms are necessary for the emergence of rhythmicity in Group 2 (the non-rhythmic group). Each condition was run with 5 independent seeds and 200 permutations for the rhythmicity test. We report the sine-fit amplitude A and the permutation p-value for Group 2.

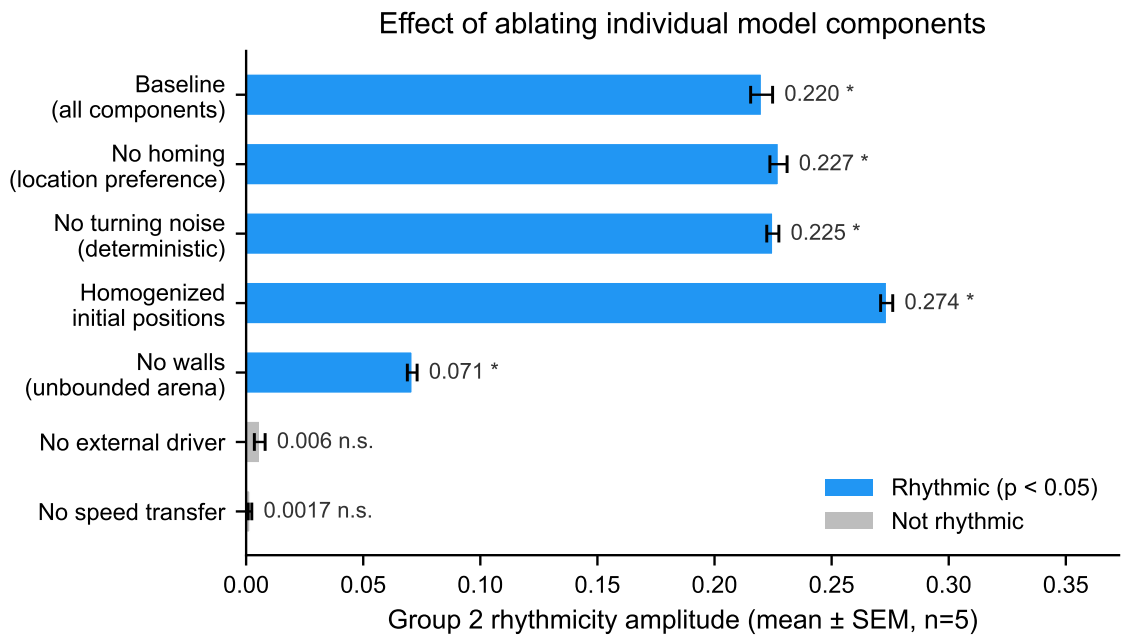


Figure S4: Effect of ablating individual model components on Group 2 rhythmicity amplitude (mean ± SEM, n=5 seeds). Speed transfer is the only component whose removal eliminates Group 2 rhythmicity entirely ($A_2 \approx 0.002$, $p = 0.36$). Walls have a secondary amplifying effect, reducing A_2 from 0.223 to 0.059 (confinement increases contact rate). Homing and turning noise have negligible effects. Homogenizing initial positions slightly *increases* transfer ($A_2 = 0.234$), since mixing groups spatially produces more early inter-group contacts.

Table 10: Default parameter values used in all simulations unless otherwise noted.

Symbol	Parameter	Value	Unit
Arena			
L_x	Comb width	354	mm
L_y	Comb height	205	mm
Environment			
σ_s	Speed scale	50	–
h	Homing pull coefficient	0.1	–
γ	Speed transfer decay	0.8	–
ρ	Interaction threshold factor	0.2	–
d_{int}	Interaction threshold	$\sqrt{50} \times 0.2 \approx 1.41$	mm
Timing			
T_{day}	Steps per simulated day	400	steps
T_{sim}	Total simulation duration	1200	steps
Group 1 (rhythmic)			
μ_1	Home position	$(-175, -100)$	mm
μ_1^{fwd}	Mean forward speed	0.05	mm/step
σ_1^{fwd}	Std forward speed	0.02	mm/step
$v_{\min,1}$	Minimum forward speed	-0.05	mm/step
μ_1^{turn}	Mean turning rate	0.0	rad/step
σ_1^{turn}	Std turning rate	0.5	rad/step
Group 2 (non-rhythmic)			
μ_2	Home position	$(0, 0)$	mm
μ_2^{fwd}	Mean forward speed	0.02	mm/step
σ_2^{fwd}	Std forward speed	0.01	mm/step
$v_{\min,2}$	Minimum forward speed	-0.05	mm/step
μ_2^{turn}	Mean turning rate	0.0	rad/step
σ_2^{turn}	Std turning rate	0.5	rad/step
Agent counts			
N_g	Agents per group	$\lfloor 500 \times d \rceil$	–
	(density factor d)	1.0 (default)	–

11.7 Supplementary Note 2: Statistical verification of spatial and temporal correlations

To verify the correlations between age, spatial position, and circadian rhythmicity reported in the main text, we performed ordinary least-squares (OLS) regressions, with each occupied comb position contributing one observation. All models were highly significant ($p < 0.001$), and the sign of every relationship is consistent across the two colonies.

Distance from entrance vs. Age

Confirming the spatial sorting of bees (Fig. 1d), age significantly predicts distance from the entrance in both colonies: older bees are found closer to the entrance (negative coefficient in 2016 and 2019 alike).

Table 11: OLS Regression: Distance from entrance ~ Age

Dataset	Coef.	Std.Err.	t	P> $\|t\|$	R^2	N
2016 (Colony A)						
Intercept	359.29	0.83	432.9	<0.001	0.194	82,124
Age	-7.23	0.05	-140.5	<0.001		
2019 (Colony B)						
Intercept	327.01	0.86	379.2	<0.001	0.087	87,557
Age	-6.04	0.07	-91.4	<0.001		

Circadian Phase vs. Age

Confirming that older bees have an earlier phase (peak activity earlier in the day), we found a consistent negative correlation between phase and age in both years.

Table 12: OLS Regression: Circadian Phase ~ Age

Dataset	Coef.	Std.Err.	t	P> $\|t\|$	R^2	N
2016 (Colony A)						
Intercept	12.80	0.01	1464.4	<0.001	0.024	82,124
Age	-0.024	0.001	-45.2	<0.001		
2019 (Colony B)						
Intercept	13.68	0.01	2519.8	<0.001	0.037	87,557
Age	-0.024	0.000	-57.8	<0.001		

Distance vs. Circadian Phase

Confirming Fig. 1e, bees further from the entrance have a significantly later phase.

Table 13: OLS Regression: Distance from entrance ~ Circadian Phase

Dataset	Coef.	Std.Err.	t	P> |t|	R^2	N
2016 (Colony A)						
Intercept	-9.29	4.45	-2.1	0.037	0.040	82,124
Phase	20.78	0.36	58.1	<0.001		
2019 (Colony B)						
Intercept	-1172.32	5.60	-209.4	<0.001	0.426	87,557
Phase	106.54	0.42	255.0	<0.001		

Distance vs. Rhythm Strength (R^2)

Confirming Fig. 1e, bees further from the entrance have a significantly weaker circadian rhythm (lower R^2 of the cosinor fit).

Table 14: OLS Regression: Distance from entrance ~ Rhythm Strength (R^2)

Dataset	Coef.	Std.Err.	t	P> |t|	R^2	N
2016 (Colony A)						
Intercept	336.52	0.41	816.2	<0.001	0.437	82,124
R^2	-590.15	2.34	-252.7	<0.001		
2019 (Colony B)						
Intercept	393.56	0.59	663.0	<0.001	0.435	87,557
R^2	-758.73	2.92	-259.5	<0.001		

Linear Model of Speed Change

To verify that speed difference predicts speed change (confirming the trend in Fig. 2a using a continuous model rather than bins), we fitted an Ordinary Least Squares (OLS) model.

Table 15: OLS Regression: Speed Change ~ (Speed Other - Speed Self)

Variable	Coef.	Std.Err.	t	P> |t|	R^2	N
Intercept	0.080	0.000	202.6	<0.001	0.078	13,289,384
Speed diff.	0.154	0.000	648.9	<0.001		

11.8 Parameter Sensitivity

To assess the robustness of the speed-transfer model, we performed single-parameter sweeps over the three main tunable parameters: agent density d, day duration T_{day}, and decay rate γ. In each sweep

Table 16: Ablation study: effect of removing individual model components on rhythmicity transfer (mean ± SEM across n=5 independent seeds). A_1 and A_2 are the fitted sinusoidal amplitudes of the Group 1 (forager) and Group 2 (hive bee) mean speed time series, respectively. $\Delta\phi$ is the phase lag of Group 2 relative to Group 1 in degrees, and p_2 is the permutation-test p-value for Group 2 rhythmicity (200 permutations). Simulations used density factor d=1.0, day duration T_{day}=100 steps, and decay rate γ=0.8. Speed transfer is the sole essential mechanism; walls provide secondary amplification by confining agents and increasing contact rates.[a]

Condition	A_1	A_2	$\Delta\phi$ (°)	p_2
Baseline (all components)	1.25 ± 0.00	0.220 ± 0.005	14.8 ± 0.6	<0.01
No speed transfer	1.25 ± 0.00	0.0017 ± 0.0008	—	0.65
No homing (location preference)	1.25 ± 0.00	0.227 ± 0.004	12.7 ± 1.3	<0.01
No walls (unbounded arena)	1.25 ± 0.00	0.071 ± 0.002	20.2 ± 4.3	<0.01
No turning noise	1.25 ± 0.00	0.225 ± 0.003	14.8 ± 0.6	<0.01
No external driver	0.01 ± 0.00	0.006 ± 0.002	—	0.36
Homogenized initial positions	1.24 ± 0.00	0.274 ± 0.003	8.6 ± 1.0	<0.01
No homing + no speed transfer	1.25 ± 0.00	0.0025 ± 0.0002	—	0.31
No walls + no speed transfer	1.25 ± 0.00	0.0022 ± 0.0003	—	0.42
No walls + no homing	1.25 ± 0.00	0.069 ± 0.004	7.7 ± 1.6	<0.01
No walls, homing, or speed transfer	1.25 ± 0.00	0.0031 ± 0.0005	—	0.21

[a]Phase shift omitted where Group 2 rhythmicity was not significant ($p \geq 0.05$).

one parameter is varied while all others are held at their baseline values (d=1.0, T_{day}=100 steps, γ=0.8); each condition is evaluated with n=5 independent seeds and 200 permutations.

Summary of parameter effects

Density (d). The density factor scales the number of agents in the arena and thereby controls the availability of interaction partners. Higher density increases the rate of inter-group contacts, which leads to faster build-up of rhythmicity in Group 2 (larger A_2) and correspondingly shorter propagation delays (smaller $\Delta\phi$). Both effects are monotonic: A_2 increases approximately linearly from 0.07 at d=0.25 to 0.37 at d=3.0, while $\Delta\phi$ decreases from ∼17° to ∼9° (Fig. S6).

Day duration (T_{day}). The day duration sets the number of simulation steps per light–dark cycle. A longer cycle provides more samples per day, making the temporal resolution finer, but it also means the exponential decay γ is applied more times per day—the effective per-day memory is $\gamma^{T_{\text{day}}}$, which becomes negligibly small for large T_{day} (e.g. $0.8^{400} \approx 10^{-39}$). In practice, amplitude is largely invariant to day duration ($A_2 \approx 0.22$ for $T_{\text{day}} \geq 75$), confirming that the steady-state transfer signal does not depend on temporal resolution. By contrast, phase lag scales inversely with T_{day}: $\Delta\phi \approx 31°$ at T_{day}=50 down to $\Delta\phi \approx 2°$ at T_{day}=800 (Fig. S7). This relationship arises because the propagation delay is approximately constant in absolute time steps; as the cycle length increases, this fixed delay represents a progressively smaller fraction of the full 360° cycle.

Decay rate (γ). The decay rate implements a leaky memory of past speed transfers: at each time step, the accumulated speed-transfer signal is multiplied by γ, so γ=0 corresponds to no memory (only instantaneous interactions) and values approaching 1 retain nearly all past information. Amplitude increases monotonically with γ ($A_2 \approx 0.06$ at γ=0 to $A_2 \approx 0.30$ at γ=0.9), because longer memory allows the rhythmic signal to accumulate across multiple interactions (Fig. S8). Crucially, rhythmicity transfer remains significant ($p < 0.01$) even at γ=0, demonstrating that the basic mechanism does not require temporal integration. Phase lag also increases substantially with γ, approximately tripling from $\Delta\phi \approx 7^\circ$ at γ=0 to $\Delta\phi \approx 21^\circ$ at γ=0.9. This is because the accumulated memory acts as inertia: at high γ, an agent's speed is heavily influenced by interactions from many previous steps, so it takes longer for the current phase of the driving rhythm to override the historical signal, delaying the response of Group 2.

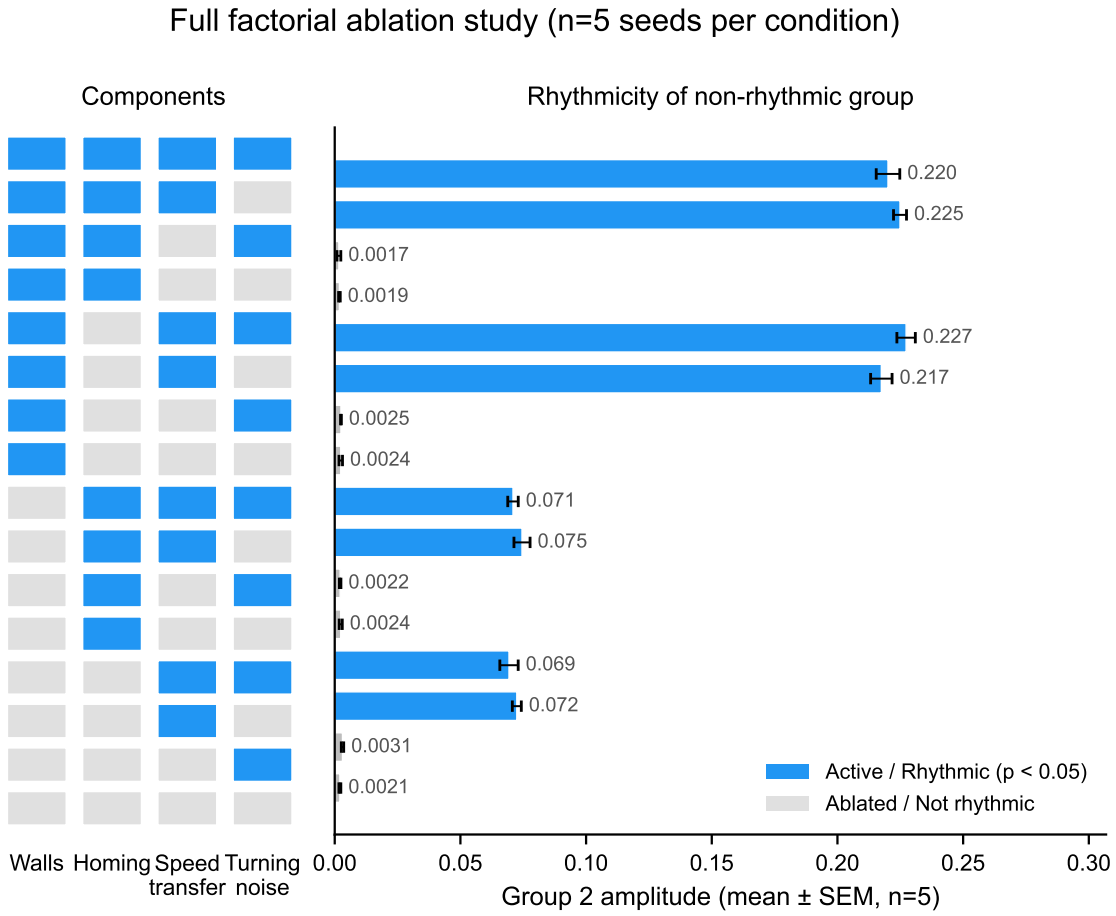


Figure S5: Full 2^4 factorial ablation over four structural components (n=5 seeds per condition). Left: blue squares indicate the component is active; grey indicates it has been ablated. Right: Group 2 amplitude. Whenever speed transfer is ablated (grey in the third column), Group 2 amplitude collapses to near zero regardless of other component states.

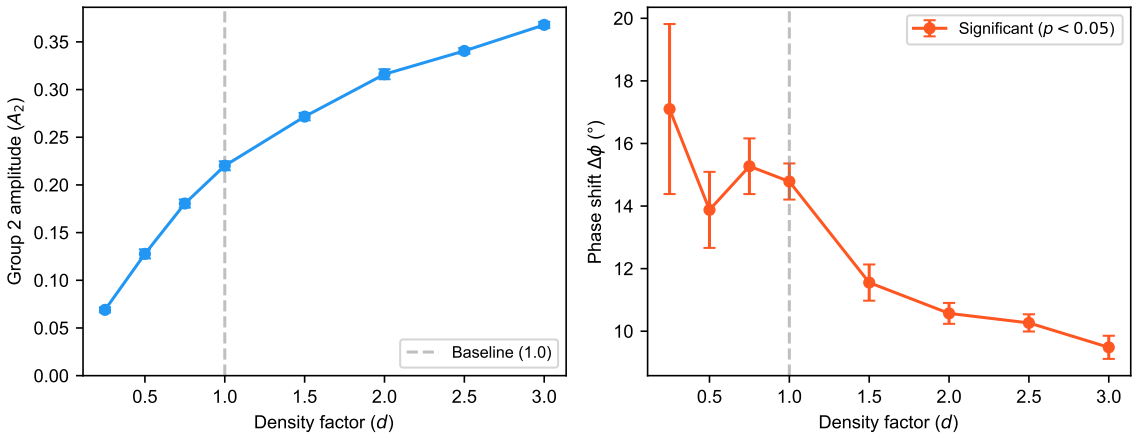


Figure S6: Sensitivity to agent density ($d \in [0.25, 3.0]$, T_{day}=100, γ=0.8). **Left:** Group 2 amplitude A_2 increases monotonically with density. **Right:** Phase lag $\Delta\phi$ decreases with density, as more available interaction partners accelerate signal propagation. Dashed line marks the baseline (d=1.0).

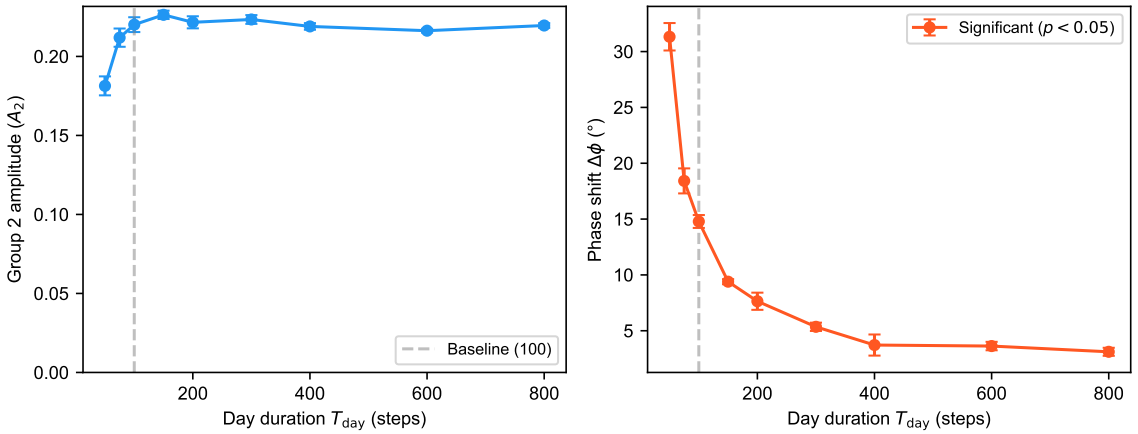


Figure S7: Sensitivity to day duration ($T_{\text{day}} \in [50, 800]$ steps, d=1.0, γ=0.8). **Left:** Amplitude is nearly constant for $T_{\text{day}} \geq 75$. **Right:** Phase lag scales inversely with day duration, consistent with a fixed propagation delay measured in steps. Dashed line marks the baseline (T_{day}=100).

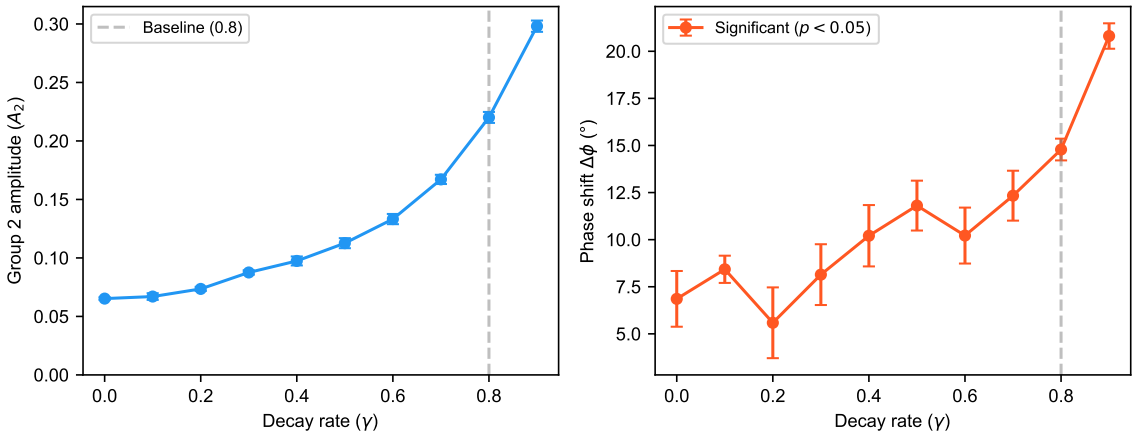


Figure S8: Sensitivity to decay rate ($\gamma \in [0.0, 0.9]$, d=1.0, T_{day}=100). **Left:** Amplitude increases with γ as longer memory amplifies the accumulated signal. Importantly, transfer is significant ($p < 0.01$) even at γ=0. **Right:** Phase lag increases approximately threefold from γ=0 to γ=0.9, reflecting inertia from accumulated memory. Dashed line marks the baseline (γ=0.8).

11.9 Supplementary Note 3: Robustness of rhythm detection and multiple-testing correction

Generalized linear model versus OLS cosinor

Because movement speed is strictly positive and right-skewed, the normal-error assumption of the ordinary least-squares (OLS) cosinor is not fully met. We therefore refit every bee's rhythm with a generalized linear model (GLM; Gamma family, log link), as recommended in review (Doyle et al., 2022), on the same 10-minute-binned speed data, and compared it with the OLS cosinor used in the main text (colony B, $n = 3{,}525$ bees with sufficient data). The two approaches agree on rhythmic classification for 99.6% of bees (94.2% vs. 94.4% classified rhythmic at $p < 0.05$; disagreeing for only 15 bees) and yield near-identical acrophase estimates (median difference 0.07 h, i.e. ∼4 min; 90th percentile 0.20 h). The GLM thus confirms the OLS results, and our conclusions do not depend on the distributional assumptions of the linear cosinor.

Multiple-testing correction

When rhythms are fit for thousands of bees, the per-bee p-values can be adjusted for multiple testing. Applying a Benjamini–Hochberg (FDR) correction to the per-bee cosinor p-values decreases the proportion of significantly rhythmic bees only modestly, and the increase of rhythmicity with age (Fig. 1c) is preserved in both colonies (Table 17). Likewise, the 6×6 interaction tests comparing measured against null-model speed changes remain significant after correction (Table 18), because the per-cell sample sizes are very large.

Table 17: Proportion of significantly rhythmic bee-days per age cohort ($p < 0.05$), before and after Benjamini–Hochberg (FDR) correction. The increase of rhythmicity with age (Fig. 1c) is preserved.

	Colony A (2016)		Colony B (2019)	
Age cohort	raw	FDR	raw	FDR
0–4 days	31%	26%	66%	61%
5–9 days	56%	51%	69%	65%
10–14 days	58%	53%	69%	65%
15–19 days	68%	63%	73%	69%
20–24 days	70%	64%	79%	77%
25–29 days	74%	69%	75%	72%
35–39 days	76%	71%	81%	80%
overall	66%	61%	71%	67%

Table 18: Interaction tests remain significant after Benjamini–Hochberg (FDR) correction: number of significant cells (of 36; Welch test, $\alpha = 0.05$) before and after correction.

Colony	Metric	Uncorrected	FDR-corrected
A (2016)	age	36/36	36/36
A (2016)	phase	36/36	36/36
A (2016)	R^2	36/36	36/36
A (2016)	start speed	34/36	34/36
B (2019)	age	36/36	36/36
B (2019)	phase	36/36	36/36
B (2019)	R^2	36/36	36/36
B (2019)	start speed	35/36	35/36

11.10 Supplementary Note 4: Spectral estimate of the rhythm period

Rather than assuming a 24 h period, we estimated the dominant period of each bee's movement-speed rhythm directly using Lomb–Scargle spectral analysis (appropriate for the unevenly sampled time series). Averaging the periodograms across all significantly rhythmic bees (n = 2,700; colony B) yields a single dominant peak at 23.9 h with its expected 12 h harmonic and no competing periodicities (Figure S9a). The per-bee peak periods are tightly centred just below 24 h (median 23.9 h, interquartile range 23.8–24.2 h; Figure S9b), confirming that a 24 h period accurately describes the population rhythm while showing the "close to but not exactly 24 h" character characteristic of a circadian clock. The spectral analysis used the full per-bee time series and was performed on colony B; colony A shows the same age–rhythmicity structure in the supplementary figures (Figs. S1–S2).

We additionally examined whether the estimated period depends on age. A weak relationship is present (younger bees tend towards slightly longer periods), but it is partly confounded by record length: shorter time series bias the period estimate, and younger bees have shorter records. Controlling for record span (whose own effect on the estimate, −0.030 h per day, is comparable in magnitude), a modest age effect persists (−0.017 h per day of age, $p < 10^{-6}$). We therefore regard the age-dependence of the period as suggestive rather than conclusive.

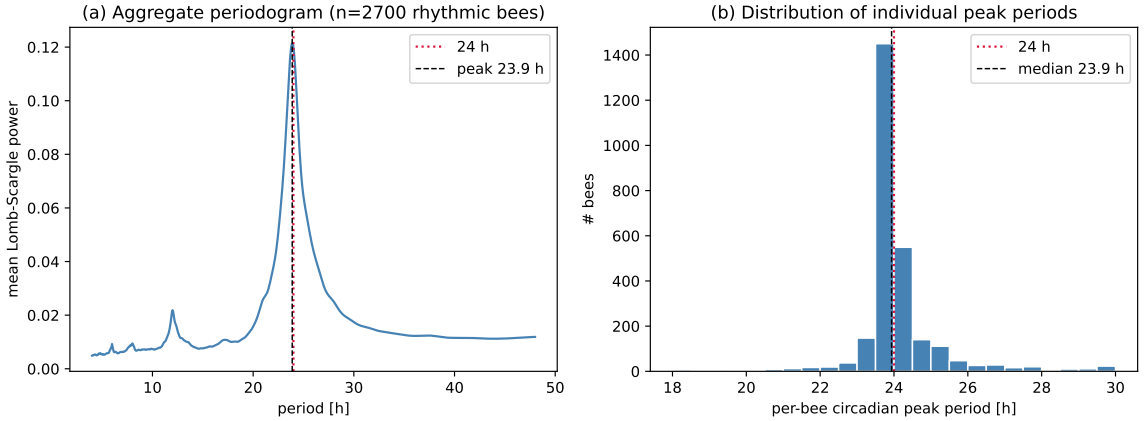


Figure S9: Spectral estimate of the rhythm period (colony B, n = 2,700 rhythmic bees). **(a)** Mean Lomb–Scargle periodogram: a single dominant peak at 23.9 h with a 12 h harmonic and no competing periodicities. **(b)** Distribution of individual peak periods, tightly centred just below 24 h (median 23.9 h).

11.11 Supplementary Note 5: Day/night comparison of the spatial age distribution

Reviewer 2 asked whether the distribution of bee age across distance from the entrance (Fig. 1d) changes between day and night. Figures 1d,e are computed from the start positions of recorded interactions, so we split exactly this substrate by time of day: each interaction contributes both partners' start position and age, interaction times are converted to local (Berlin) time, and we compare day (09:00–18:00) with night (21:00–06:00), the day/night windows used elsewhere in this study. We report this for colony B, the colony of the main text (81.0/50.1 million day/night position samples).

The age–distance profile keeps the same overall shape by day and by night (Figure S10): median age falls from the oldest bees near the entrance (≈ 20 days) to a young brood-nest core (≈ 7 days), and the young core is essentially unchanged between day and night. The day/night difference is confined to the older-bee zone near the entrance, which extends further into the nest during the day: at 136 comb-units from the entrance the median age is 18 days by day versus 13 at night, and at 175 units 13 versus 9. A linear model age $\sim$ distance $\times$ day/night confirms a highly significant day/night difference in the gradient (interaction term $p \ll 10^{-15}$; $n = 131$ million observations).

This modulation is expected from the underlying biology: the older, strongly rhythmic bees near the entrance are most active during the day and therefore dominate daytime interactions there, whereas at night, as their activity subsides, the interacting population is more uniformly young. The spatial age organisation—young near the entrance, youngest in the brood core—is thus a stable feature present around the clock, whose apparent amplitude is modulated by the daytime activity of the older bees. As for Fig. 1d, this measure reflects activity-weighted occupancy.

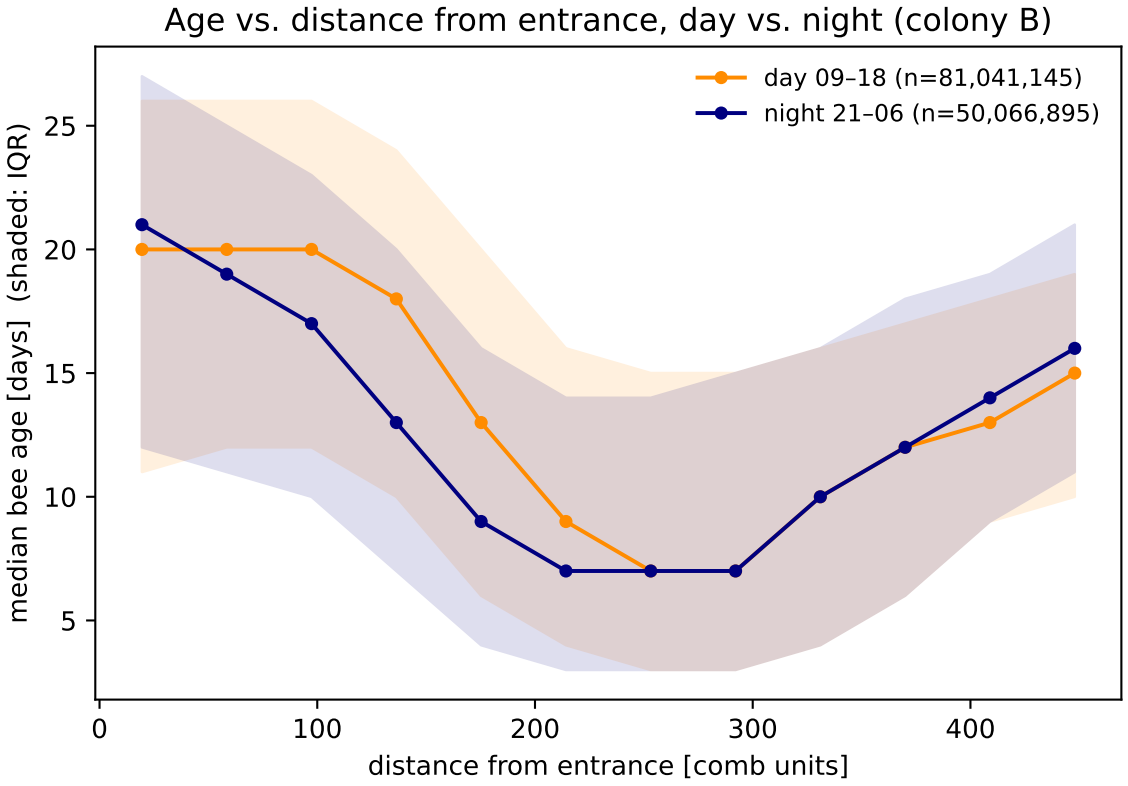


Figure S10: Day/night comparison of the spatial age distribution (Reviewer 2, colony B). Median bee age as a function of distance from the entrance, computed from interaction start positions and split into day (09:00–18:00, orange) and night (21:00–06:00, blue); shaded bands show the interquartile range. The young brood-nest core is present at the same distance by day and night; the day/night difference is concentrated in the older-bee zone near the entrance, where the age contrast is larger during the day.

11.12 Supplementary Figure: interaction-tree depth versus location and local density

We examined whether the depth of the activation-cascade trees (Fig. 3c) could reflect the local density of bees rather than distance from the entrance. Interaction-tree depth is spatially structured across the comb rather than uniform (Figure S11a), and depends only weakly on the local sampling density (Spearman $\rho \approx 0.15$; density explains ≈2% of the variance in depth; Figure S11b). The high-depth region co-locates with the nest entrance rather than with the highest-density region, so the systematic increase of tree depth towards the entrance is unlikely to be an artefact of spatially varying density. We note that the per-location depth values are spatially smoothed, so this relationship should be read as indicative; a fully density-matched null model of the cascade trees is left for future work.

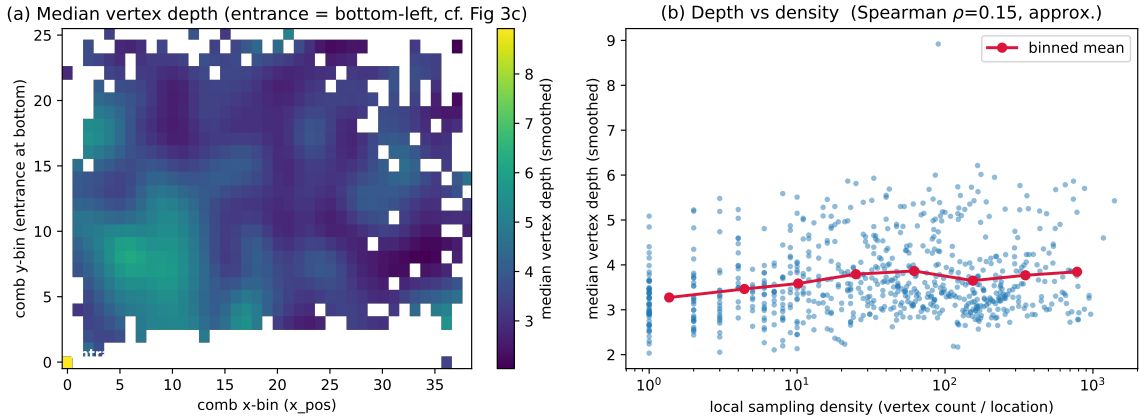


Figure S11: Interaction-tree depth is spatially structured and only weakly density-dependent. **(a)** Median vertex depth per hive location (entrance at bottom-left). **(b)** Depth vs. local sampling density (vertex count per location); the relationship is weak (Spearman $\rho \approx 0.15$).